Mediated Excitation Transfer in Equity Markets: A Protocol-Topological Theory of Narrative Bosons, Rotation Forces, and No-Trace Price Fluctuations

A Protocol-Topological Theory of Narrative Bosons, Rotation Forces, and No-Trace Price Fluctuations

Draft Installment 1 — Abstract, Reader's Guide, and Sections 1–3

Abstract

Equity markets are commonly analyzed through returns, risk factors, valuation ratios, liquidity, investor expectations, and information flow. Yet many market episodes display a structure that is not fully captured by ordinary factor exposure or random-walk descriptions. One stock becomes unusually visible; its price, volume, volatility, and attention traces intensify; market observers compress that visible event into a tradable explanation; and the resulting theme, template, or narrative is projected onto adjacent securities. The heat appears to rotate. Stock A rises, becomes a signal, emits a transferable interpretation, and Stock B becomes the next carrier.

This paper proposes a protocol-topological framework for analyzing such mediated excitation transfer in equity markets. The central object is not a stock price in isolation, but a declared stock manifold: a protocol-bound state space containing prices, abnormal returns, volume, attention, liquidity, ownership overlap, factor exposure, thematic similarity, news flow, social co-mentions, options activity, and institutional gates. Within this manifold, a short-lived price movement may be interpreted as a self-referential excitation if the price trace itself changes observer attention, search behavior, and capital routing.

The paper introduces three core concepts. First, a **Narrative Boson** is a transferable market mediator formed when investors, analysts, algorithms, media, or thematic funds compress the visible excitation of one stock into a tradable theme. Second, an **Observer-Mediated Rotation Force** is the directional pressure by which this mediator increases the abnormal response probability of another stock. Third, a **No-Trace Price Fluctuation** is a virtual-like excitation: a price movement that appears, propagates, and decays without leaving durable traces in earnings expectations, ownership structure, credit access, capital formation, policy classification, or long-term valuation regime.

The framework does not claim that equity markets literally instantiate quantum field theory. It uses quantum-style language only as a functional role grammar: field, mediator, gate, trace, invariance, curvature, and residual are treated as analytical roles, not physical substances. This follows the protocol-first discipline of Gauge Grammar, where claims must be made under declared boundary, observation rule, time window, and admissible intervention, rather than under uncontrolled metaphor.

Methodologically, the paper combines factor-adjusted residual returns, similarity graphs, event-study logic, trace-persistence tests, and LLM-assisted topology kernels. The proposed LLM layer is not a price predictor. It is a qualitative structure compiler: it declares protocol, maps the financial manifold, detects curvature, identifies candidate mediators, names possible attractors, marks bifurcation gates, and audits residual uncertainty. This follows the Runtime Kernels framework, where topology-inspired opcodes such as Boundary, Curvature, Attractor, Projection, Flow, Invariant, and Residual are valid only when each has a concrete operation and output evidence.

The central thesis is:

(0.1) Equity-market heat does not merely diffuse; it can be mediated, compressed, transferred, absorbed, damped, and ledger-tested.

The practical aim is to convert market-rotation stories into testable topology hypotheses.



Keywords

Equity markets; market microstructure; behavioral finance; narrative finance; factor residuals; thematic rotation; self-referential pricing; mediated excitation; virtual fluctuations; Gauge Grammar; LLM financial analysis; topology kernel; residual audit.

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0. Reader's Guide

This paper is written for readers in finance, economics, market structure, risk management, equity

research, quantitative strategy, behavioral finance, and AI-assisted financial analysis. It does not assume acceptance of any metaphysical theory. It does not propose a literal physical theory of markets. It proposes a disciplined analogy converted into an operational financial research framework.

The motivating observation is simple.

In many market episodes, a stock does not rise alone. Its rise becomes a signal. The signal becomes a story. The story becomes a search pattern. The search pattern identifies nearby securities. Capital then rotates toward those securities. This produces a common market phrase:

“The heat moved to the next stock.”

This paper asks whether that phrase can be made analytically precise.

The answer proposed here is yes, but only under protocol.

A useful model must declare:

(0.2) $P = (B, \Delta, h, u)$.

Where:

Symbol	Meaning in this paper
B	Boundary: which stock universe is being studied
Δ	Observation rule: which traces count as excitation or response
h	Horizon: lag window for emission, propagation, absorption, and decay
u	Admissible correction: market, sector, factor, liquidity, and event adjustments

Without this declaration, the language of “narrative bosons,” “rotation forces,” or “virtual excitations” would be only metaphor. With protocol, it becomes a research grammar.

The attached finance OpCode document gives a suitable methodological foundation. It maps Manifold, Coordinate, Boundary, Curvature, Flow, Gradient, Attractor, Basin, Bifurcation, Singularity, Projection, Invariant, Holonomy, Residual, Compression, and Phase-lock into finance-specific analytical operations. It also presents the “Finance Topology Kernel” as a way to compile market reality into a runtime analysis structure rather than relying on loose commentary.

This paper should therefore be read at three levels.

First, it is a **finance article** about factor-adjusted abnormal returns, cross-stock spillovers, attention, liquidity, and thematic rotation.

Second, it is a **market-structure article** about how price traces become observer inputs, how observer inputs affect capital routing, and how capital routing feeds back into price.

Third, it is an **AI-assisted analysis article** about how LLMs may help compile qualitative financial topology before numerical validation.

The central caution is:

(0.3) Functional analogy ≠ literal identity.

A stock is not a particle. A narrative is not a physical boson. A price move is not a quantum event. The claim is narrower:

(0.4) Certain financial roles are structurally similar to field, mediator, gate, trace, residual, and force roles.

The value of the framework must be judged by whether it improves diagnosis, test design, residual honesty, and cross-frame analysis.

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1. Introduction — From Random Walks to Mediated Excitations

1.1 The standard view and its blind spot

Modern finance has powerful tools for describing price behavior. Random walks, efficient-market models, factor models, volatility models, liquidity models, behavioral anomalies, and network contagion frameworks all capture important parts of market reality. A stock's return can be decomposed into market beta, sector exposure, factor exposure, liquidity shocks, earnings news, and residual noise.

A familiar decomposition is:

(1.1) $r_{i,t} = \alpha_{i,t-1} + \beta_{i,t-1}\, r_{m,t} + \beta^{s}_{i,t-1}\, r^{s},t + \beta^{f}_{i,t-1}\, F_t + \varepsilon_{i,t}$.

Where $r_{i,t}$ is stock i's return, $r_{m,t}$ is the market return, r^{s},t is the sector return, F_t is a vector of factor returns, and $\varepsilon_{i,t}$ is the residual.

In ordinary analysis, $\varepsilon_{i,t}$ is often treated as idiosyncratic return, noise, unexplained movement, firm-specific information, or alpha candidate. This paper proposes that part of $\varepsilon_{i,t}$ may have a more specific structure.

Some residual movements are not merely random. They are self-referential market excitations. They arise because the market observes its own price trace.

A stock rises. The rise becomes visible. The visibility attracts observers. Observers search for an explanation. The explanation becomes tradable. Other stocks are then reclassified under that explanation. Those stocks rise. The original movement has therefore altered the future probability distribution of adjacent securities.

In compact form:

(1.2) PriceTrace_A → ObserverAttention_A → NarrativeCompression_θ → Projection_B → AbnormalResponse_B.

This is not simple correlation. It is mediated excitation transfer.

1.2 Why equity markets are especially suitable

Commodities often have strong physical anchors: production cost, storage cost, substitution, inventory, transportation, and consumption. Their short-term fluctuations may sometimes resemble virtual-like disturbances around a slower fundamental attractor.

Equities are different. Equity prices are claims on expected future cash flows, but those expectations are themselves influenced by price. A rising stock can improve financing capacity, employee morale, acquisition currency, media visibility, analyst coverage, and investor confidence. A falling stock can damage these channels. This makes equity prices unusually self-referential.

The earlier *Peaks, Traps, and the Trinity* argument is relevant here because it identifies finance as the domain where price is recursively tied to its own expected future value. In that framework, financial price is not merely a parameter; it is a recursive fixed point inside expectation.

This paper adopts that insight but applies it to cross-stock transfer. It asks:

If price can recursively affect itself, can one stock's price trace affect another stock through a transferable mediator?

The proposed answer is yes.

1.3 A basic example

Suppose Stock A rises sharply after an AI-related catalyst. The move is visible in price, volume, options activity, news coverage, and social attention. Investors then ask:

Which other companies benefit from the same theme?

This search may identify Stock B, which has not yet moved but shares a supply-chain, revenue, theme, ownership, or ETF-basket relation with A. Stock B then rises, not because its own fundamentals have immediately changed, but because market observers project the same theme onto it.

The process can be summarized as:

(1.3) A excitation → AI narrative boson → B absorption → B excitation.

The term "boson" is used here functionally. It means a mediator that carries interaction between market objects. It does not mean a physical particle.

A more finance-native description is:

(1.4) A abnormal trace → tradable theme → similarity search → B abnormal flow.

This paper keeps both descriptions because the first captures the structural analogy and the second preserves the finance interpretation.

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2. Conceptual Definitions

2.1 Stock manifold

A **stock manifold** is the declared state space in which mediated excitation transfer is studied.

(2.1) $\mathcal{M}_P$ = {Price, Volume, Volatility, Attention, Liquidity, Theme, Ownership, Options, Fundamentals, Risk | P}.

Under protocol P, each stock is represented not only by price but by a multi-dimensional coordinate vector:

(2.2) $X_{i,t} = (p_{i,t}, r_{i,t}, v_{i,t}, a_{i,t}, l_{i,t}, \theta_{i,t}, o_{i,t}, q_{i,t}, f_{i,t}, \kappa_{i,t})$.

Where:

Symbol	Meaning
$p_{i,t}$	price
$r_{i,t}$	return
$v_{i,t}$	volume or turnover
$a_{i,t}$	attention measure
$l_{i,t}$	liquidity

Symbol	Meaning
$\theta_{i,t}$	theme vector or narrative embedding
$o_{i,t}$	ownership / ETF / fund overlap structure
$q_{i,t}$	options activity
$f_{i,t}$	fundamental revision state
$\kappa_{i,t}$	risk or curvature proxy

The manifold is protocol-bound because different studies require different coordinates. A short-squeeze study needs short interest and borrow cost. A sector-rotation study needs ETF overlap and factor exposure. A supply-chain transfer study needs customer-supplier data. A narrative-contagion study needs text embeddings and attention metrics.

The Gauge Grammar discipline is essential here: bad protocol produces bad diagnosis, and proxies are compiled estimates rather than revealed essence.

2.2 Excitation

A **local excitation** is an abnormal movement in a stock's residual price-attention-volume state after admissible corrections.

First remove common field movement:

(2.3) $\varepsilon_{i,t} = r_{i,t} - E_{t-1}[r_{i,t} \mid \text{market, sector, factors, liquidity, known events}]$.

Then define excitation:

(2.4) $E_i(t) = w_1 z(\varepsilon_{i,t}) + w_2 z(\text{VolumeSurprise}_{i,t}) + w_3 z(\text{Attention}_{i,t}) + w_4 z(\text{OptionActivity}_{i,t})$.

Where $z(\cdot)$ denotes standardization and w_1, w_2, w_3, w_4 are declared weights.

The purpose is not to create a universal excitation measure. The purpose is to make the excitation definition explicit.

A stock is considered excited when:

(2.5) $E_i(t) > \theta_E$.

Where θ_E is the declared excitation threshold.

2.3 Narrative boson

A **Narrative Boson** is a transferable market mediator formed by compressing a visible excitation into a tradable interpretation.

(2.6) $NB_{A,\theta,t} = \text{Compress}(\text{Trace}_{A,t} \mid \theta, P)$.

Where:

Symbol	Meaning
$NB_{A,\theta,t}$	narrative boson emitted by Stock A under theme θ
$\text{Trace}_{A,t}$	visible price-volume-attention-news trace of A
θ	theme frame
P	declared protocol

In plain finance language:

A narrative boson is what the market extracts from A's move and uses to search for B.

Examples include:

A-side excitation	Narrative boson
AI stock rally	AI infrastructure theme
uranium stock rally	nuclear energy theme
semiconductor rally	data-center capex theme
defence rally	geopolitical security theme
meme-stock rally	short-squeeze template
bank selloff	deposit-flight / duration-risk template
GLP-1 rally	obesity-drug ecosystem theme

The term "boson" is justified only if it performs an analytical role: it must mediate a measurable relation between A and B.

Thus:

(2.7) ValidNarrativeBoson ⇔ MediatorRole ∧ Transferability ∧ ObservableTrace ∧ TestableAbsorption.

If it cannot be observed, transferred, or tested, it is not a useful analytical mediator.

2.4 Absorption

A stock B **absorbs** a narrative boson when its probability of abnormal response rises after A's excitation, conditional on coupling.

(2.8) $\text{Absorb}_B(NB_A,\theta,t) \propto g_{A,B,\theta}(t) \cdot U_B(t) \cdot L_B(t) \cdot V_B(t)$.

Where:

Term	Meaning
$g_{A,B,\theta}(t)$	theme-specific coupling between A and B
$U_B(t)$	unsaturation: B has not yet fully moved
$L_B(t)$	liquidity / tradability
$V_B(t)$	visibility / discoverability

The absorption concept prevents the model from treating every related stock as equally affected. A stock may be thematically similar but illiquid, already crowded, fundamentally denied, or invisible to the relevant investor base. In that case, the mediator may not be absorbed.

2.5 Rotation force

The **Observer-Mediated Rotation Force** from A to B is the directional pressure generated when A's excitation, a transferable mediator, and B's absorptive capacity align.

The core expression is:

(2.9) $F_{A\to B,\theta}(t,k) = g_{A,B,\theta}(t) \cdot E_A(t) \cdot U_B(t) \cdot L_B(t) \cdot M_{A,B,\theta}(t) - D_B(t)$.

Where:

Symbol	Meaning

Symbol	Meaning
$F_{A\to B,\theta}(t,k)$	rotation force from A to B over lag k under theme θ
$g_{A,B,\theta}(t)$	coupling strength
$E_A(t)$	A-side excitation
$U_B(t)$	B-side unsaturation
$L_B(t)$	B-side liquidity
$M_{A,B,\theta}(t)$	mediator strength
$D_B(t)$	damping

This is not a pricing equation. It is a structural equation for hypothesis formation.

The empirical question is:

(2.10) Does $F_{A\to B,\theta}(t,k)$ predict $\varepsilon_{B,t+k}$ after admissible correction?

2.6 Virtual and real excitations

A price fluctuation is **virtual-like** when it does not leave durable ledger trace.

(2.11) $\text{Virtual}_B \Leftrightarrow \Delta p_B \neq 0 \wedge \Delta \text{Ledger}_B \approx 0$.

A price fluctuation is **real** when it changes the future admissible path of the stock.

(2.12) $\text{Real}_B \Leftrightarrow \Delta p_B \neq 0 \wedge \Delta \text{Ledger}_B \neq 0$.

Ledger variables may include:

Ledger Layer	Possible evidence
Earnings ledger	analyst EPS revisions, revenue revisions, margin forecasts
Ownership ledger	persistent institutional inflows, ETF weight changes
Credit ledger	spread change, refinancing access, capital-raise ability
Narrative ledger	durable analyst reclassification, persistent thematic inclusion
Policy ledger	subsidy, regulation, export control, legal classification
Valuation ledger	persistent multiple expansion or compression
Corporate-action ledger	issuance, M&A, capex, hiring, strategic pivot

A no-trace fluctuation may still be tradable. It may be large. It may be visible. But if it does not alter the future ledger, it remains virtual-like.

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3. Protocol-Topological Finance

3.1 Why protocol comes first

Financial analysis easily becomes unstable because the object changes across frames.

A stock can be cheap under earnings, expensive under free cash flow, attractive under a growth frame, dangerous under a credit frame, safe under accounting classification, and fragile under liquidity stress. The same observed price is not the same object under every protocol.

This is why the paper begins with protocol:

(3.1) $P = (B, \Delta, h, u)$.

Where:

Component	Meaning
B	Boundary
Δ	Observation rule
h	Time or state window
u	Admissible intervention or correction family

This follows the protocol-first logic of Gauge Grammar, where a bounded observer must declare boundary, observation rule, time window, and admissible intervention before making a system claim. The framework also warns that local correctness does not guarantee governed closure.

In this paper:

(3.2) Claim_P = Interpret(Σ_P | B, Δ, h, u).

Where Σ_P is the trace observed under protocol P.

A statement such as “heat rotated from A to B” is incomplete unless it declares:

- Which A and B?
- Which universe contains them?
- Which time horizon defines transfer?
- Which common factors were removed?
- Which mediator is claimed?
- Which evidence would falsify the transfer?
- Which residual remains?

3.2 Boundary

Boundary defines the market object.

Possible boundaries include:

Boundary type	Example
Sector boundary	semiconductor stocks
Theme boundary	AI infrastructure stocks
ETF boundary	stocks in a thematic ETF
Supply-chain boundary	data-center power chain
Ownership boundary	stocks co-held by the same funds
Factor boundary	high-beta growth stocks
Event boundary	post-earnings movers
Squeeze boundary	high short-interest low-float stocks

The boundary determines what counts as adjacent.

In one protocol, Stock B may be adjacent to A because it is in the same industry. In another, it may be adjacent because both are in the same ETF. In another, it may be adjacent because retail traders classify both as “meme squeeze” candidates.

Thus:

(3.3) Adjacency_P(A,B) depends on B, Δ, h, and u.

3.3 Observation rule

The observation rule defines what counts as excitation and response.

Possible observations include:

Observation family	Variables
Price	abnormal return, residual return, drawdown reversal
Volume	volume surprise, turnover
Attention	media, social, search, analyst coverage
Options	call volume, put-call skew, implied volatility
Ownership	ETF flow, institutional flow, short interest
Narrative	co-mentions, topic embeddings, theme tags
Fundamentals	earnings revision, margin expectation
Credit	CDS, bond spread, refinancing access

A minimal observation rule might use only returns and volume. A richer rule might add attention, options, and news embeddings.

The paper's preferred observation rule is:

(3.4) Δ = {residual return, volume surprise, attention surprise, mediator trace, ledger persistence}.

This rule is sufficient to distinguish mere price movement from mediated excitation and no-trace decay.

3.4 Horizon

The horizon h determines the life cycle of the excitation.

Short horizons may detect intraday attention transfer. Weekly horizons may detect thematic rotation. Monthly horizons may detect real rerating.

For this framework, h has four sub-windows:

(3.5) h = (h_emit, h_prop, h_absorb, h_decay).

Where:

Sub-window	Meaning
h_emit	time window in which A emits visible excitation
h_prop	time window in which mediator spreads
h_absorb	lag in which B responds
h_decay	window used to test virtual decay or real trace

Example:

(3.6) h = (1 day, 3 days, 5 days, 20 days).

This means A's excitation is detected over one day, mediator propagation is observed over three days, B response is measured over five days, and ledger persistence is tested over twenty days.

3.5 Admissible corrections

Admissible corrections prevent false transfer claims.

At minimum:

(3.7) u = {market beta, sector beta, factor exposure, liquidity shock, known direct news}.

A richer protocol may include:

Correction	Purpose
market beta	remove broad market movement
sector beta	remove sector-wide movement
factor beta	remove style and macro exposures
liquidity adjustment	remove liquidity-wide shock
volatility adjustment	remove risk-regime effect
direct news filter	exclude B's independent catalyst
earnings-window filter	avoid event contamination
ETF-flow adjustment	separate passive basket flow from narrative effect

The goal is not to remove everything. The goal is to avoid mistaking common shocks for mediated transfer.

The residual object is:

(3.8) ResidualTransfer_A→B = ObservedResponse_B − BestAdmissibleCorrection_B.

If residual transfer remains after admissible correction, it becomes a candidate mediated excitation.

This aligns with the Gauge Grammar formula:

(3.9) Curvature_P = ObservedGap_P − BestLocalCorrection_P.

In finance terms, curvature is what remains when all valid local frame corrections have been attempted.

3.6 Residual as research object

Residual is not merely error.

In this framework, residual may be:

- missing data;
- unmodeled factor;
- hidden liquidity route;
- theme mediator;
- cross-frame curvature;
- unpriced optionality;
- crowding;

- institutional constraint;
- observer disagreement.

The Runtime Kernels framework makes residual audit central: a valid kernel must preserve objective, boundary, output contract, and residual audit; a good opcode must have operation and output evidence.

For this paper, residual has a precise role:

(3.10) Residual_P = what remains unexplained after protocol-declared correction and before metaphorical interpretation.

Only after residual is identified should the model ask whether it reflects a narrative boson, liquidity mediator, attention gradient, or no-trace fluctuation.

3.7 The first methodological rule

The first methodological rule of the paper is:

(3.11) No protocol, no boson.

A claimed mediator must be attached to:

- declared stock universe;
- declared observation rule;
- declared lag horizon;
- declared corrections;
- declared coupling measure;
- declared trace-persistence test.

Otherwise, the claim is only a story.

The second rule is:

(3.12) No trace test, no virtuality.

A fluctuation cannot be called no-trace unless future ledger variables have been examined.

The third rule is:

(3.13) No coupling measure, no force.

A force claim requires an explicit measure of A-B coupling.

Together:

(3.14) ValidTransferClaim_P ⇔ ProtocolDeclared ∧ ExcitationDetected ∧ MediatorIdentified ∧ CouplingMeasured ∧ ResponseTested ∧ ResidualAudited.

This is the operational foundation for the rest of the paper.

Transition to the Next Part

The next sections should build the measurable model:

Section 4 — Equity-Manifold Construction and Factor-Neutral Residuals
Section 5 — Narrative Bosons as Transferable Market Mediators
Section 6 — Coupling, Absorption, and Observer-Mediated Rotation Force

Draft Installment 2 — Sections 4–6

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4. Equity-Manifold Construction and Factor-Neutral Residuals

4.1 Why the raw price chart is not the research object

A stock price chart is not yet a stock manifold. It is only one projection of a higher-dimensional financial object. A stock simultaneously exists as:

- a price process;
- a balance-sheet claim;
- an earnings expectation;
- a liquidity object;
- an index constituent;
- an options-underlying;
- a media object;
- a social-attention object;
- a legal-security object;
- a narrative carrier;
- a governance and accounting trace.

This is why raw co-movement is insufficient. If Stock A rises and Stock B rises one day later, that does not yet prove mediated excitation transfer. Both may simply share market beta, sector exposure, factor loading, earnings-cycle sensitivity, liquidity shock, or index-flow pressure.

The first task is therefore to remove the ordinary shared field.

Let raw return be:

(4.1) $r_{i,t} = \Delta \log(P_{i,t})$.

A basic admissible correction is:

(4.2) $r_{i,t} = \alpha_{i,t-1} + \beta_{m,i,t-1}\, r_{m,t} + \beta_{s,i,t-1}\, r_{s,t} + \beta_F_{,i,t-1} \cdot F_t + \varepsilon_{i,t}$.

Where:

Symbol	Meaning
$r_{i,t}$	return of stock i

Symbol	Meaning
r_m,t	broad market return
r_s,t	relevant sector or industry return
F_t	factor vector
ε_i,t	factor-neutral residual return

The residual ε_i,t is the first candidate location of local excitation.

This step is important because the paper is not trying to identify ordinary market exposure. It is trying to identify **residual mediated excitation** after admissible frame correction. This matches the finance OpCode idea that financial analysis should first define the market object, select variables, detect hidden tensions, classify regimes, and audit residual risk rather than jumping directly from price movement to interpretation.

4.2 Rolling estimation and anti-look-ahead discipline

The correction must be estimated without using future information.

Therefore:

(4.3) $\beta_i,t-1 = \text{Estimate}(\beta_i \mid \text{data} \le t-1)$.

Not:

(4.4) $\beta_i = \text{Estimate}(\beta_i \mid \text{full sample})$.

The second form contaminates past interpretation with future structure. If the aim is to model how observers at time t could have interpreted the market, then only information available before t should be used.

A proper empirical design should therefore use:

Estimation method	Use
rolling beta	adapts to changing exposure
expanding beta	stable long-history estimate
state-space beta	time-varying latent exposure
regime beta	exposure conditional on market state
ex-sector market beta	avoids using the sector to explain itself

For sector or theme studies, the broad market factor should ideally exclude the target sector when the sector is large enough to affect the index.

(4.5) $r_m,\text{ex-s},t$ = market return excluding target sector s.

Otherwise, one may partly use Stock B's own field to explain Stock B, creating self-cancellation.

4.3 The equity manifold

After admissible factor adjustment, each stock can be represented as a coordinate vector in a protocol-declared manifold.

(4.6) $X_i,t = (\varepsilon_i,t, \nu_i,t, a_i,t, \ell_i,t, \sigma_i,t, q_i,t, \theta_i,t, o_i,t, c_i,t, f_i,t)$.

Where:

Coordinate	Meaning
$\varepsilon_{i,t}$	residual return
$v_{i,t}$	volume surprise
$a_{i,t}$	attention surprise
$\ell_{i,t}$	liquidity condition
$\sigma_{i,t}$	realized or implied volatility
$q_{i,t}$	options activity
$\theta_{i,t}$	theme vector
$o_{i,t}$	ownership / ETF / fund overlap
$c_{i,t}$	coupling map to other stocks
$f_{i,t}$	fundamental revision state

The equity manifold under protocol P is:

(4.7) $\mathcal{M}_P = \{X_{i,t} \mid i \in B, t \in h, \text{observed under } \Delta \text{ and corrected under } u\}$.

This definition prevents a common error: treating “the market” as a single object. Finance is a price field, liquidity network, legal-contract system, accounting ledger system, narrative-expectation field, risk-transfer machine, and multi-observer collapse system. The attached OpCode document makes this explicit and argues that the first use of OpCodes in finance is to force the analyst to declare which financial world is being analyzed.

4.4 Residual excitation score

Once $\varepsilon_{i,t}$ is obtained, local excitation can be measured as a composite residual event.

(4.8) $E_i(t) = w_1 z(\varepsilon_{i,t}) + w_2 z(v_{i,t}) + w_3 z(a_{i,t}) + w_4 z(q_{i,t}) + w_5 z(\sigma_{i,t})$.

Where:

Term	Meaning
$z(\varepsilon_{i,t})$	standardized abnormal residual return
$z(v_{i,t})$	standardized volume surprise
$z(a_{i,t})$	standardized attention surprise
$z(q_{i,t})$	standardized options activity
$z(\sigma_{i,t})$	standardized volatility surprise

A stock is considered locally excited when:

(4.9) $E_i(t) > \theta_E$.

This does not yet mean that the excitation is mediated or meaningful. It only means that Stock i has produced a visible trace strong enough to enter the analysis.

The distinction is:

(4.10) Excitation ≠ Transfer.

A stock can be excited without transmitting anything. A stock can transmit without causing absorption. A stock can cause absorption without leaving durable ledger trace.

4.5 Excitation half-life

The next question is whether the excitation decays.

Define a residual price path:

(4.11) $Y_{i,t} = \Sigma_{s \le t} \varepsilon_{i,s}$.

A virtual-like residual excitation should display finite half-life. A simple mean-reverting approximation is:

(4.12) $\Delta Y_{i,t+1} = -\kappa_i Y_{i,t} + \eta_{i,t+1}$.

If $\kappa_i > 0$, the half-life is approximately:

(4.13) $HL_i = \ln(2) / \kappa_i$.

A no-trace excitation should satisfy:

(4.14) $HL_i < K$ and $TracePersistence_i \approx 0$.

Where K is the declared maximum decay horizon.

This helps separate short-lived heat from structural repricing.

4.6 Residual curvature

Not all residuals are noise. Some residuals are curvature.

In this paper:

(4.15) $Curvature_P = ObservedResponse_P - BestAdmissibleCorrection_P$.

If the response remains after market, sector, factor, liquidity, and direct-news corrections, the residual becomes a candidate curvature point.

This is where the OpCode method becomes useful. In the Runtime Kernel dictionary, Curvature is not decorative language. It means detecting nonlinear tension or contradiction, while Residual means reporting the unresolved remainder.

For the present article, residual curvature may mean:

- unmodeled theme transfer;
- hidden ownership overlap;
- attention contagion;
- ETF basket pressure;
- relative-value catch-up;
- options gamma channel;
- old-leader memory;
- analyst-frame reclassification;

- liquidity-routing effect.

The next section introduces the mediator that can make such residual curvature intelligible.

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5. Narrative Bosons as Transferable Market Mediators

5.1 Why a mediator is needed

If Stock A rises and Stock B later rises, there are three possibilities.

First, both may have moved because of the same external factor.

Second, Stock B may have moved because of its own independent catalyst.

Third, Stock A's visible excitation may have changed the observer field, causing Stock B to be reinterpreted, searched, bought, and repriced.

Only the third case is mediated excitation transfer.

The mediator is necessary because Stock A does not physically push Stock B. The causal bridge is interpretive, institutional, and tradable.

A possible transfer chain is:

(5.1) A excitation → visible trace → observer compression → transferable mediator → B projection → B response.

The mediator is called a **Narrative Boson** when the transferable bridge is primarily thematic or semantic.

5.2 Definition of Narrative Boson

A Narrative Boson is defined as:

(5.2) $NB_{A,\theta,t} = Compress(Trace_{A,t} \mid \theta, P)$.

Where:

Symbol	Meaning
$NB_{A,\theta,t}$	narrative boson emitted from Stock A under theme θ at time t
$Trace_{A,t}$	observed excitation trace of A
θ	theme frame
P	declared protocol

The compression operation converts a high-dimensional event into a tradable search object.

Examples:

Visible trace	Compressed mediator
"A rises after AI demand surprise"	AI infrastructure boson
"A squeezes on high short interest"	short-squeeze template boson
"A rallies after policy subsidy"	policy-support boson
"A jumps on defence orders"	geopolitical-security boson

Visible trace	Compressed mediator
“A rerates on energy scarcity”	energy-security boson
“A moves after GLP-1 demand”	metabolic-health ecosystem boson

The mediator is not merely a story. It must be capable of changing capital routing.

Thus:

(5.3) Valid NB ⇔ VisibleTrace ∧ TradableTheme ∧ TransferableClassification ∧ AbsorptionCandidateSet.

If there is no candidate set of securities that can absorb the mediator, the narrative is not yet a market boson.

5.3 Narrative boson versus news

A Narrative Boson is not the same as news.

News may be local. A Narrative Boson is transferable.

For example:

- “Company A beat earnings” may be local news.
- “AI server demand is accelerating” may be a transferable mediator.
- “A bank has deposit outflow” may be local stress.
- “Regional bank duration risk is underpriced” may be a transferable mediator.

The distinction is:

(5.4) News_A = information about A.

(5.5) NB_A,θ = compressed interpretation that can be projected beyond A.

This is why some earnings reports move only one stock, while others move an entire cluster.

A report becomes bosonic when it is not merely information but an interaction carrier.

5.4 Emission conditions

A stock does not emit a strong narrative boson merely by moving. Emission requires visibility and compressibility.

Emission strength can be written as:

(5.6) Emit_A,θ(t) = E_A(t) · C_θ(Trace_A,t) · V_A(t) · R_θ(t).

Where:

Term	Meaning
E_A(t)	A-side excitation
C_θ(Trace_A,t)	compressibility of A’s trace into theme θ
V_A(t)	visibility of A to relevant observers

Term	Meaning
$R_{\theta}(t)$	current market receptivity to theme θ

The theme must already be intelligible or quickly made intelligible. A very obscure explanation may fail to propagate even if A moves sharply.

Thus:

(5.7) $StrongEmission_{A,\theta} \Leftrightarrow HighExcitation \wedge HighVisibility \wedge HighCompressibility \wedge HighThemeReceptivity$.

This explains why some small-cap moves disappear while others become templates.

5.5 Propagation channels

A Narrative Boson propagates through channels.

Possible channels include:

Channel	Mechanism
media	articles, headlines, financial TV
analyst reports	peer read-through, target revisions
social platforms	crowd attention and meme replication
ETF baskets	passive and thematic flow routes
fund ownership	common holders rebalance across peers
options market	call buying, gamma dynamics, volatility surface
screeners	abnormal volume, momentum, short-interest filters
supply-chain maps	customer-supplier implication
factor models	style and risk bucket reclassification
broker notes	“beneficiary basket” creation

Propagation strength is:

(5.8) $M_{A,B,\theta}(t) = ChannelIntensity_{A,B,\theta}(t) \cdot ObserverOverlap_{A,B}(t)$.

Where ObserverOverlap means the degree to which the same investor communities observe both A and B.

A theme cannot transfer if the relevant observers do not see the candidate absorber.

5.6 Absorber set

For each emitted mediator, define an absorber set:

(5.9) $\mathcal{B}_{A,\theta,t} = \{B \mid g_{A,B,\theta}(t) > \theta_g \text{ and } Eligibility_{B,\theta}(t) = 1\}$.

Eligibility may require:

- business-model relevance;
- theme match;
- tradeability;

- non-saturation;
- enough liquidity;
- no immediate disconfirming catalyst;
- inclusion in the observer's search universe.

For example, after an AI infrastructure rally, the absorber set may include power equipment, cooling, networking, memory, data-center landlords, and semiconductor equipment names. But not every company with an AI press release should qualify.

The protocol must prevent narrative overreach.

5.7 Decay of the mediator

Narrative bosons decay.

A simple decay form is:

(5.10) $NB_{A,\theta,t+k} = NB_{A,\theta,t} \cdot \exp(-\delta_\theta k)$.

Where δ_θ is the theme decay rate.

But theme decay is not constant. It depends on reinforcement.

(5.11) $\delta_{\theta,t} = \delta_0 - Reinforcement_{\theta,t} + Disconfirmation_{\theta,t} + Saturation_{\theta,t}$.

Where:

Term	Effect
Reinforcement	earnings, news, price continuation, analyst support
Disconfirmation	weak data, failed read-through, management denial
Saturation	crowded trade, valuation stress, fatigue
δ_0	natural attention decay

If reinforcement is strong enough, the mediator can become durable. It stops being merely virtual and begins to reshape the market's ledger.

5.8 Narrative boson and phase-lock

If many observers accept the same mediator, phase-lock occurs.

(5.12) $PhaseLock_\theta(t) = Agreement(Investors, Analysts, Media, Funds, Algorithms \mid \theta)$.

When $PhaseLock_\theta$ is high, multiple agents move in rhythm. The attached finance OpCode mapping treats phase-lock as herding, index rebalancing, central-bank narrative alignment, or synchronized selling.

In equity rotation:

(5.13) High $PhaseLock_\theta \rightarrow$ stronger propagation, lower skepticism, faster absorption.

But high phase-lock also increases fragility. If the theme breaks, many observers may reverse simultaneously.

Thus, phase-lock is both amplifier and risk.

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6. Coupling, Absorption, and Observer-Mediated Rotation Force

6.1 From correlation to coupling

Correlation measures co-movement.

Coupling measures admissible transfer.

Two stocks may be correlated because they share market beta. That is not enough. A mediated-transfer model needs a specific A-B channel.

Define coupling:

(6.1) $g_{A,B,\theta}(t) = Similarity_{A,B,\theta}(t) \cdot Route_{A,B}(t) \cdot ObserverOverlap_{A,B}(t)$.

Where:

Term	Meaning
$Similarity_{A,B,\theta}$	similarity under theme θ
$Route_{A,B}$	institutional or liquidity route connecting A and B
$ObserverOverlap_{A,B}$	degree of shared investor attention

This coupling is theme-specific. The same two stocks may be strongly coupled under one frame and weakly coupled under another.

For example:

(6.2) $g_{A,B,AI} \neq g_{A,B,Defence} \neq g_{A,B,RateSensitivity}$.

This is a gauge-like point: changing the interpretive frame changes the connection map.

6.2 Coupling measures

Possible proxies for $g_{A,B,\theta}$ include:

Coupling dimension	Possible proxy
business similarity	revenue-segment overlap
theme similarity	text embedding similarity
industry similarity	classification distance
supply-chain relation	customer-supplier link
ETF overlap	shared ETF constituents
fund overlap	common institutional holders
analyst overlap	same analyst coverage universe
news co-mention	co-occurrence in articles
social co-mention	co-occurrence in posts
options crowding	similar call/put pressure
factor exposure	shared factor loading

Coupling dimension	Possible proxy
historical activation	prior co-rally episodes

A composite coupling measure may be:

(6.3) $g_{A,B,\theta}(t) = \Sigma_j \lambda_j \cdot g_{j,A,B,\theta}(t)$.

Where j indexes coupling channels and λ_j are declared weights.

A more flexible model may allow time-varying weights:

(6.4) $g_{A,B,\theta}(t) = \Sigma_j \lambda_{j,t} \cdot g_{j,A,B,\theta}(t)$.

This is important because market frames change. During one period, ETF overlap may dominate. During another, social co-mention may dominate. During a crisis, credit linkage may dominate.

6.3 Unsaturation

A candidate absorber must not already be saturated.

Define unsaturation:

(6.5) $U_B(t) = 1 - Sat_B(t)$.

Where $Sat_B(t)$ may depend on:

- recent abnormal return;
- valuation rerating;
- volume crowding;
- options crowding;
- short-term attention exhaustion;
- fund ownership concentration;
- analyst optimism saturation.

A simple form:

(6.6) $Sat_B(t) = \sigma(a_1 CumRet_{B,t-n:t} + a_2 Attention_{B,t} + a_3 ValuationZ_{B,t} + a_4 Crowding_{B,t})$.

Where σ is a logistic transform.

Then:

(6.7) High $U_B(t) \rightarrow$ B is more available as a next carrier.

This captures the familiar market phrase:

“Stock A has already moved; Stock B has not moved yet.”

But the framework formalizes that phrase as unsaturation.

6.4 Liquidity and tradeability

A stock may be thematically perfect but untradeable.

Liquidity enters as:

(6.8) $L_B(t) = f(\text{turnover}_{B,t}, \text{spread}_{B,t}, \text{depth}_{B,t}, \text{borrow}_{B,t}, \text{options}_{B,t})$.

A high liquidity absorber can attract institutional rotation. A low liquidity absorber may attract speculative heat but can also become unstable.

Thus liquidity has two roles:

(6.9) Liquidity as capacity: high L_B supports larger absorption.

(6.10) Illiquidity as convexity: low L_B can amplify price response for small flows.

A more complete model may split liquidity:

(6.11) $L_B(t) = \text{Capacity}_B(t) + \text{Convexity}_B(t)$.

Where capacity allows flow absorption, and convexity creates explosive movement.

6.5 Damping

Damping prevents excitation transfer.

Define damping:

(6.12) $D_B(t) = d_1\text{ValuationResistance}_{B,t} + d_2\text{LiquidityFriction}_{B,t} + d_3\text{FundamentalDenial}_{B,t} + d_4\text{RiskLimit}_{B,t} + d_5\text{Crowding}_{B,t}$.

Examples:

Damping source	Interpretation
valuation resistance	B already expensive
liquidity friction	hard to trade efficiently
fundamental denial	market rejects theme fit
risk limits	funds cannot buy more
crowding	too many already positioned
negative catalyst	B has independent bad news
accounting concern	reported story not trusted
credit stress	equity theme blocked by solvency risk

Damping is crucial because mediated transfer is not automatic.

The market may try to project A's theme onto B, but B may reject the projection.

In this case:

(6.13) High mediator strength + high damping → failed absorption.

Failed absorption is itself informative. It may reveal that the market's initial theme is too broad, too late, or already exhausted.

6.6 Rotation force

We can now restate the core force equation:

(6.14) $F_{A\to B,\theta}(t,k) = g_{A,B,\theta}(t) \cdot E_A(t) \cdot U_B(t) \cdot L_B(t) \cdot M_{A,B,\theta}(t) - D_B(t)$.

Where $M_{A,B,\theta}(t)$ is mediator strength.

A positive force means:

(6.15) $F_{A\to B,\theta}(t,k) > 0 \rightarrow$ increased probability of abnormal residual response in B.

A negative or weak force means:

(6.16) $F_{A\to B,\theta}(t,k) \leq 0 \rightarrow$ no expected mediated absorption.

The empirical response model is:

(6.17) $\varepsilon_{B,t+k} = c + \beta_1 E_{A,t} + \beta_2 g_{A,B,\theta,t} + \beta_3 E_{A,t} g_{A,B,\theta,t} + \beta_4 U_{B,t} + \beta_5 L_{B,t} + \beta_6 M_{A,B,\theta,t} - \beta_7 D_{B,t} + \eta_{B,t+k}$.

The main mediated-transfer coefficient is β_3.

(6.18) $\beta_3 > 0$ supports mediated excitation transfer.

A stronger model tests the full interaction:

(6.19) $\varepsilon_{B,t+k} = c + \beta_F F_{A\to B,\theta}(t,k) + \text{Controls} + \eta_{B,t+k}$.

If $\beta_F > 0$ and remains significant after controls, the model supports the existence of observer-mediated rotation force.

6.7 Force direction

The force does not act in physical space. It acts in an attention-capital manifold.

A direction can be represented as:

(6.20) $\text{Direction}_{A\to B,\theta} = \nabla_B \text{ProjectionValue}(B \mid NB_{A,\theta,t})$.

That is:

> capital tends to move toward the candidate absorber whose projection value rises most after A's excitation.

A simplified projection value:

(6.21) $PV_{B,\theta}(t) = \text{Fit}_{B,\theta}(t) + \text{UpsideImagination}_{B,\theta}(t) + \text{LiquidityAccess}_B(t) - \text{Saturation}_B(t) - \text{Damping}_B(t)$.

Then:

(6.22) $\text{Direction} = \text{argmax}_B PV_{B,\theta}(t)$.

This captures the market question:

"Where is the next best carrier of this theme?"

Rotation force is the pressure created by that search.

6.8 Absorption probability

Instead of modeling return directly, one may model the probability that B becomes excited after A.

(6.23) $\Pr(E_B(t+k) > \theta_E \mid E_A(t), \theta) = \sigma(\gamma_0 + \gamma_1 F_A\to B,\theta(t,k) + \gamma_2 \text{Controls})$.

Where σ is a logistic transform.

This formulation is useful because mediated transfer may be nonlinear. Once B crosses a visibility threshold, attention and order flow may increase sharply.

The transfer is therefore not merely additive. It may be thresholded:

(6.24) Absorption_B occurs if $F_A\to B,\theta(t,k) > \Theta_B$.

Where Θ_B is B's activation threshold.

The threshold may fall when:

- market theme receptivity is high;
- B is already on watchlists;
- B has rising social mentions;
- B has high short interest;
- B has options activity;
- B is included in relevant ETFs.

The threshold may rise when:

- the market is risk-off;
- B is illiquid;
- B has credibility problems;
- the theme is saturated;
- B has negative direct news.

6.9 Bifurcation gates

A mediated excitation may remain virtual or become real depending on gate events.

Possible bifurcation gates:

Gate	Meaning
earnings confirmation	B reports numbers consistent with theme
analyst inclusion	B enters beneficiary basket
ETF inclusion	B enters passive flow route
options acceleration	gamma channel amplifies move
management confirmation	company validates theme exposure
policy confirmation	subsidy / regulation supports theme

Gate	Meaning
capital access	B raises funds at favorable terms
credit response	bond or lender market confirms rerating
ownership persistence	long-term holders enter

A bifurcation is:

(6.25) Gate_B,t = 1 if event changes future admissible path of B.

The transition from virtual to real is:

(6.26) Virtual_B + Gate_B,t → Real_B.

This is one of the most important claims of the paper.

A price move becomes “real” not merely because it is large, but because it passes through a gate that writes into ledger.

6.10 Relation to Peaks and Traps

The mediated-transfer model can also generate Peaks and Traps.

A Peak occurs when price increase raises demand.

(6.27) ∂Demand_B / ∂p_B > 0 under active self-reference.

This may happen when rising price validates the theme and attracts more buyers.

A Trap occurs when the system enters a non-escape basin.

(6.28) NegativeTrace_B → lower access → worse expectation → further negative trace.

For example:

- stock price falls;
- financing becomes harder;
- market confidence deteriorates;
- analysts cut estimates;
- further price decline occurs.

The *Peaks, Traps, and the Trinity* document explicitly treats Peaks and Traps as witnesses of the incompleteness of standard systems that do not treat recursive expectations and institutional closure as primitive.

In this paper, mediated excitation transfer can be benign, virtual, peak-forming, or trap-forming.

Transfer outcome	Meaning
virtual heat	short-lived no-trace move
real rerating	durable ledger change
Peak	self-reinforcing positive demand
Trap	self-reinforcing negative basin
phase transition	theme becomes structural regime

6.11 Summary of the model

The full structure can now be summarized:

(6.29) A excitation = abnormal residual price-volume-attention trace.

(6.30) Narrative boson = compressed transferable mediator from A.

(6.31) Coupling = admissible A-B relation under theme θ.

(6.32) Absorption = B's response to mediator under coupling, unsaturation, liquidity, and damping.

(6.33) Rotation force = directional pressure in the attention-capital manifold.

(6.34) Virtuality = absence of durable ledger trace.

(6.35) Reality = durable deformation of future admissible path.

The next step is to convert these definitions into empirical classification.

Transition to the Next Part

The next installment should cover:

Section 7 — Virtual Excitation, Real Excitation, and Ledger Trace
Section 8 — Empirical Design: Event Study, Similarity Graphs, and Propagation Tests
Section 9 — LLM OpCode Kernel as Qualitative Topology Compiler

Draft Installment 3 — Sections 7–9

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7. Virtual Excitation, Real Excitation, and Ledger Trace

7.1 Why the virtual / real distinction matters

A stock can rise without becoming structurally different.

It may attract attention for one day, appear in trading screens, trigger momentum buying, and then decay. After the episode, analysts do not revise earnings, funds do not maintain ownership, credit markets do not change their view, management does not alter capital strategy, and the market no longer carries the theme into future valuation.

Such a movement was visible, tradable, and real as a transaction event. But in the framework of this paper, it was not structurally real. It did not deform the future ledger.

This paper therefore distinguishes:

(7.1) Price visibility ≠ ledger persistence.

A **virtual excitation** is a price movement that appears in the observable market trace but does not alter the future admissible path of the security.

A **real excitation** is a price movement that passes through a gate and changes the future ledger.

The distinction is not metaphysical. It is empirical. We ask:

- Did the price move persist?
- Did it alter future earnings expectations?
- Did it change institutional ownership?
- Did it improve or damage financing access?
- Did analysts reclassify the stock?
- Did the firm change behavior because of the move?
- Did the theme become part of the firm's durable valuation frame?

If the answer is mostly no, the excitation is virtual-like.

If the answer is yes, the excitation became real.

7.2 Ledger variables

A ledger is any durable record that changes future interpretation, action, constraint, or admissibility.

In finance, ledger trace is not limited to accounting books. It includes market, institutional, narrative, and legal traces.

Ledger type	Examples
Price ledger	persistent price level, support/resistance memory, long-term chart structure
Valuation ledger	durable multiple expansion, peer group change, target-price revision
Earnings ledger	EPS revisions, revenue revisions, margin revisions
Ownership ledger	institutional accumulation, ETF inclusion, long-only ownership shift
Credit ledger	bond spread change, refinancing access, rating outlook
Capital ledger	equity issuance, convertible financing, M&A currency
Narrative ledger	durable thematic classification, analyst basket inclusion
Policy ledger	subsidy, regulation, tariff, export-control classification
Corporate ledger	hiring, capex, strategic pivot, management guidance
Risk ledger	volatility regime shift, borrow cost, short-interest reset

A price move becomes structurally real when it modifies one or more of these ledgers.

Thus:

(7.2) $\Delta \text{Ledger}_{i,t:t+H} = \Delta \text{Valuation}_i + \Delta \text{Earnings}_i + \Delta \text{Ownership}_i + \Delta \text{Credit}_i + \Delta \text{Narrative}_i + \Delta \text{Capital}_i + \Delta \text{Policy}_i + \Delta \text{Risk}_i$.

A virtual-like event satisfies:

(7.3) $\Delta \text{Ledger}_{i,t:t+H} \approx 0$.

A real event satisfies:

(7.4) $\| \Delta \text{Ledger}_{i,t:t+H} \| > \theta_L$.

Where θ_L is the declared trace threshold.

7.3 Price half-life and ledger persistence

A virtual-like excitation should decay within a declared horizon.

Let $Y_{i,t}$ be the residual price path:

(7.5) $Y_{i,t} = \Sigma_{s \le t} \varepsilon_{i,s}$.

A simple mean-reversion test is:

(7.6) $\Delta Y_{i,t+1} = -\kappa_i Y_{i,t} + \eta_{i,t+1}$.

The half-life is:

(7.7) $HL_i = \ln(2) / \kappa_i$.

A no-trace virtual excitation requires both decay and ledger absence:

(7.8) $\text{Virtual}_i \Leftrightarrow HL_i < K \wedge \|\Delta \text{Ledger}_{i,t:t+H}\| \le \theta_L$.

A real excitation requires either persistence or ledger deformation:

(7.9) $\text{Real}_i \Leftrightarrow HL_i \ge K \vee \|\Delta \text{Ledger}_{i,t:t+H}\| > \theta_L$.

This distinction is important because some movements decay in price but still leave ledger trace. For example, a stock may give back its rally but still gain analyst coverage, institutional ownership, or strategic financing access. Such an event is not purely virtual.

Conversely, a price move may persist for a while due to illiquidity or technical momentum but leave no fundamental trace. It may be semi-virtual: persistent in price but weak in ledger.

7.4 Four outcome classes

Combining price persistence and ledger persistence gives four classes:

Class	Price persistence	Ledger persistence	Interpretation
Pure virtual heat	low	low	short-lived no-trace fluctuation
Technical residue	high	low	price memory without deep ledger change
Hidden real trace	low	high	price fades but institutional / narrative trace remains
Real regime shift	high	high	durable repricing and ledger deformation

This is more useful than a binary classification.

The market often contains hidden real trace. For example, a stock may spike and fade, but the episode may introduce it into a thematic investor universe. Later, another catalyst reactivates the same stock. The original excitation was not fully virtual; it left memory.

This motivates a trace-memory term:

(7.10) $\text{Memory}_{i,t+H} = f(\text{PastExcitations}_i, \text{PastNarratives}_i, \text{PastOwnership}_i, \text{PastAttention}_i)$.

This memory term becomes important for Trace-Gravity Force, where old leaders or former meme stocks attract renewed attention under similar conditions.

7.5 Virtuality ratio

For a stock universe B under protocol P, define the Virtuality Ratio:

(7.11) $VR_P = N_virtual / N_total$.

Where:

Term	Meaning
N_virtual	number of detected excitations satisfying virtual criteria
N_total	total detected excitations

A high VR_P indicates a market region where short-term heat often appears and decays without structural trace.

A low VR_P indicates a market region where excitations often become durable, either because fundamentals are changing or because the market structure is highly reflexive.

We can also define a weighted ratio:

(7.12) $WVR_P = \Sigma_j \text{Amp}_j \cdot 1\{\text{Virtual}_j\} / \Sigma_j \text{Amp}_j$.

Where Amp_j measures the magnitude of excitation j.

This is useful because many small virtual events may matter less than one large real transition.

7.6 Trace persistence function

Let $\text{Shock}_{i,t}$ be an excitation event. Define the trace persistence function:

(7.13) $TP_i(H) = \text{Response}(\Delta\text{Ledger}_{i,t:t+H} \mid \text{Shock}_{i,t})$.

In empirical work, this may be estimated by impulse response, panel regression, or event-study comparison.

Possible form:

(7.14) $\Delta\text{Ledger}_{i,t:t+H} = a + b \cdot E_i(t) + \text{Controls} + \xi_{i,t+H}$.

If $b \approx 0$, the excitation has no measurable ledger persistence.

If $b > 0$, the excitation modifies future structure.

For mediated transfer, we care about B's trace after A's excitation:

(7.15) $TP_A\to B,\theta(H) = \text{Response}(\Delta\text{Ledger_B},t:t+H \mid F_A\to B,\theta(t,k))$.

Then:

(7.16) VirtualTransfer_A→B ⇔ ε_B,t+k ≠ 0 ∧ TP_A→B,θ(H) ≈ 0.

(7.17) RealTransfer_A→B ⇔ ε_B,t+k ≠ 0 ∧ TP_A→B,θ(H) > θ_TP.

7.7 Peaks, Traps, and the virtual-real transition

Self-referential excitation can produce two dangerous structures: Peaks and Traps.

A Peak arises when price increase itself increases demand:

(7.18) $\partial \text{Demand}_i / \partial p_i > 0$.

This can occur when a rising price validates the story, attracts attention, improves perceived legitimacy, and increases buying pressure. The *Peaks, Traps, and the Trinity* article describes Peaks as cases where higher price fuels further demand, especially in status goods and bubbles, and links this to the recursive role of expectations in finance.

A Trap arises when negative price movement worsens future feasibility:

(7.19) $p_i\downarrow \rightarrow$ capital access$\downarrow \rightarrow$ confidence$\downarrow \rightarrow$ estimate revision$\downarrow \rightarrow p_i\downarrow$.

This is a non-escape basin. The same source describes Traps as recursive collapses where expectations and access barriers prevent recovery rather than mere frictions.

The virtual-real transition can therefore happen in either direction:

Transition	Mechanism
virtual → Peak	short-lived heat becomes self-validating demand
virtual → Trap	short-lived negative shock triggers constraint closure
virtual → real rerating	mediator receives fundamental confirmation
virtual → decay	no confirmation; heat dissipates

This gives a more nuanced classification:

(7.20) ExcitationOutcome ∈ {VirtualDecay, TechnicalResidue, Peak, Trap, RealRerating, RegimeTransition}.

7.8 The ledger gate

The critical question is whether the excitation passes a ledger gate.

Define:

(7.21) $\text{Gate}_{i,t} = 1$ if event changes future admissible path of stock i.

Examples of ledger gates include:

- earnings confirmation;
- analyst reclassification;
- ETF or index inclusion;
- financing event;

- regulatory classification;
- supply-chain contract;
- credit rating change;
- institutional ownership shift;
- management guidance change;
- liquidity regime shift.

Then:

(7.22) $\text{Virtual}_i + \text{Gate}_{i,t} = \text{Real}_i$.

This is the operational meaning of “becoming real.”

A movement is not real merely because it happened. It becomes real when it enters a ledger that constrains future interpretation or action.

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8. Empirical Design: Event Study, Similarity Graphs, and Propagation Tests

8.1 Research objective

The empirical goal is to test whether Stock A’s excitation predicts Stock B’s abnormal residual response through an identifiable mediator and coupling channel.

The core empirical hypothesis is:

(8.1) H1: $E_A(t) \times g_{A,B,\theta}(t)$ predicts $\varepsilon_{B,t+k}$ after admissible corrections.

A stronger hypothesis adds mediator evidence:

(8.2) H2: $E_A(t) \times g_{A,B,\theta}(t) \times M_{A,B,\theta}(t)$ predicts $\varepsilon_{B,t+k}$.

The trace hypothesis is:

(8.3) H3: Some mediated responses are no-trace virtual excitations; others become real ledger events.

8.2 Step 1 — Define the universe

The first empirical step is to declare boundary B.

Possible universes:

Universe	Suitable question
semiconductor stocks	AI / data-center heat rotation
clean-energy stocks	policy and subsidy theme transfer
defence stocks	geopolitical-security mediator
regional banks	deposit-risk and rate-risk transfer
biotech stocks	trial-result and platform-read-through transfer

Universe	Suitable question
meme stocks	short-squeeze template propagation
commodity-linked equities	commodity price mediator transfer
thematic ETF constituents	passive and narrative basket rotation

The boundary determines both possible emitters and possible absorbers.

(8.4) B = {stocks eligible for emission and absorption under protocol P}.

8.3 Step 2 — Build the similarity graph

Construct a graph G_P where nodes are stocks and edges are coupling strengths.

(8.5) G_P = (V, E, W_θ).

Where:

Symbol	Meaning
V	stocks
E	possible relationships
W_θ	theme-specific edge weights

For each pair A,B:

(8.6) W_A,B,θ = g_A,B,θ.

Possible edge components:

(8.7) g_A,B,θ = λ_1ThemeSim_A,B,θ + λ_2ETFOverlap_A,B + λ_3FundOverlap_A,B + λ_4NewsCoMention_A,B + λ_5SupplyChain_A,B + λ_6FactorSimilarity_A,B + λ_7HistoricalActivation_A,B.

This graph is not fixed. It changes with theme θ and market regime.

A bank-stock graph during deposit panic differs from a bank-stock graph during rate-cut optimism. A semiconductor graph during AI capex differs from one during inventory correction.

Thus:

(8.8) G_P(t, θ) is time-varying and frame-dependent.

8.4 Step 3 — Detect A-side excitation

A-side excitation is detected using the composite score from Section 4:

(8.9) E_A(t) = w_1z(ε_A,t) + w_2z(ν_A,t) + w_3z(a_A,t) + w_4z(q_A,t) + w_5z(σ_A,t).

An event begins when:

(8.10) E_A(t) > θ_E.

One may also classify excitation direction:

(8.11) Sign_E(A,t) = sign(ε_A,t).

Positive excitation may emit opportunity mediators. Negative excitation may emit risk mediators.

Examples:

Excitation sign	Possible mediator
positive	AI growth, policy support, squeeze template
negative	solvency concern, deposit flight, margin compression
positive then reversal	exhaustion template
negative then recovery	relief-rally template

8.5 Step 4 — Identify mediator trace

Mediator trace can be detected from text, flow, and behavior.

Possible evidence:

Mediator evidence	Proxy
news theme	topic model, embedding cluster, headline terms
social theme	co-mentioned tickers, hashtags, forum topics
analyst read-through	peer notes, basket reports
ETF route	shared thematic ETF holdings
options route	similar call-volume spike
fund route	common ownership and rotation flow
search behavior	increase in searches for “next X”
screen route	high short interest, low float, momentum filters

Define mediator strength:

(8.12) $M_A,B,\theta(t) = m_1 NewsCoTheme_A,B,\theta(t) + m_2 SocialCoAttention_A,B,\theta(t) + m_3 ETFRoute_A,B(t) + m_4 AnalystPeerRoute_A,B(t) + m_5 OptionsRoute_A,B(t)$.

A valid mediator should be observed between A’s excitation and B’s response.

Timing matters:

(8.13) $Trace_A(t) \rightarrow M_A,B,\theta(t:t+k_1) \rightarrow Response_B(t+k_2)$.

This helps distinguish mediated transfer from simultaneous common shock.

8.6 Step 5 — Measure B response

B response is measured by future residual return:

(8.14) $\varepsilon_B,t+k = r_B,t+k - E_t-1[r_B,t+k \mid market, sector, factors, liquidity, known\ B\text{-}news]$.

Then estimate:

(8.15) $\varepsilon_B,t+k = c + \beta_1 E_A,t + \beta_2 g_A,B,\theta,t + \beta_3 E_A,t g_A,B,\theta,t + \beta_4 M_A,B,\theta,t + \beta_5 U_B,t + \beta_6 L_B,t - \beta_7 D_B,t + \eta_B,t+k$.

The key term is β_3 or, in the full interaction version:

(8.16) $\varepsilon_B,t+k = c + \beta_FF_A \rightarrow B,\theta(t,k) + Controls + \eta_B,t+k$.

Where:

(8.17) $F_{A\to B,\theta}(t,k) = g_{A,B,\theta}(t) \cdot E_A(t) \cdot U_B(t) \cdot L_B(t) \cdot M_{A,B,\theta}(t) - D_B(t)$.

If $\beta_F > 0$, the data support the rotation-force hypothesis.

8.7 Step 6 — Compare against placebo pairs

To avoid overfitting, compare high-coupling pairs against placebo pairs.

For each A, construct:

Pair type	Definition
high-coupling pair	B with high $g_{A,B,\theta}$
low-coupling pair	B with low $g_{A,B,\theta}$
random pair	B randomly selected from same universe
sector-control pair	B in same sector but low theme match
theme-control pair	B theme-related but low liquidity route
time-placebo pair	same A-B pair at non-event dates

Expected result:

(8.18) Response(high coupling) > Response(placebo).

If all pairs respond similarly, the result is likely common market or sector movement, not mediated transfer.

8.8 Step 7 — Trace-persistence test

After B responds, test whether the response is virtual or real.

Define:

(8.19) $\Delta Ledger_{B,t:t+H} = Ledger_{B,t+H} - Ledger_{B,t}$.

Estimate:

(8.20) $\Delta Ledger_{B,t:t+H} = a + b \cdot F_{A\to B,\theta}(t,k) + Controls + \xi_{B,t+H}$.

If:

(8.21) $\beta_F > 0$ in price response and $b \approx 0$ in ledger response,

then the transfer is virtual-like.

If:

(8.22) $\beta_F > 0$ and $b > 0$,

then the transfer has become real.

This is the heart of the no-trace price fluctuation test.

8.9 Step 8 — Classify the topology card

Each episode can be summarized as a topology card.

Template:

```
Episode:
Emitter A:
Emission date:
Observed excitation:
Candidate mediator:
Absorber set:
Coupling evidence:
B response:
Damping:
Bifurcation gates:
Ledger trace:
Classification:
Residual:
```

Possible classifications:

Classification	Criteria
Virtual heat transfer	B responds, then decays; no ledger trace
Technical residue	B response persists in price only
Hidden trace	B price fades but ledger trace remains
Real rerating	B response and ledger trace persist
Peak	B demand rises because price rises
Trap	negative transfer closes future access
Failed absorption	mediator present but B does not respond
False mediator	response explained by common factor

This output format makes the model auditable.

8.10 Robustness checks

A serious empirical study should include:

Robustness check	Purpose
rolling beta	avoid look-ahead bias
ex-sector market factor	avoid self-cancellation
event exclusion	remove direct B catalysts
placebo dates	test random timing
placebo pairs	test random adjacency
multiple horizons	identify propagation window
alternative coupling weights	test graph sensitivity
alternative excitation thresholds	test signal stability
alternative ledger variables	test virtual / real classification
clustered standard errors	handle cross-sectional dependence
out-of-sample period	test generality

The aim is not to prove a universal law immediately. The aim is to test whether the mediated-transfer structure has empirical content beyond narrative storytelling.

9. LLM OpCode Kernel as Qualitative Topology Compiler

9.1 Why numerical methods are not enough

Numerical methods can identify patterns:

- abnormal returns;
- co-movement;
- factor exposure;
- volatility regime;
- network clusters;
- event-study effects;
- attention spikes;
- topic embeddings.

But numerical methods do not automatically name the structure.

A model may show that B responds after A when coupling is high. But it may not explain whether the structure resembles:

- liquidity routing;
- narrative propagation;
- relative-value catch-up;
- old-leader memory;
- ETF basket pressure;
- squeeze template transfer;
- factor crowding;
- accounting-projection gap;
- credit-equity frame split.

This is where an LLM OpCode layer may help. It does not replace measurement. It compiles qualitative topology.

The attached document states the relationship clearly: numerical methods discover measured patterns, while LLM OpCodes compile semantic-financial structures; together they produce topology-aware finance diagnosis.

9.2 LLM as structure compiler, not price oracle

The LLM should not be asked:

“Will Stock B go up?”

It should be asked:

“Given the evidence, what candidate topology structures could explain the observed mediated transfer, and what numerical tests would distinguish them?”

This difference is decisive.

A bad LLM use:

```
Predict the next stock that will rise after A.
```

A better LLM use:

```
Run as Finance Topology Kernel.
Declare protocol.
Map stock manifold.
Identify excitation, mediator, coupling, absorber, damping, bifurcation, residual.
Output candidate topology cards and required validation tests.
```

The Runtime Kernels framework explicitly states that the Skill should be a semantic compiler rather than a prompt decorator, and that its transformation is RawRequirement → IntentStructure → KernelIR → ExecutablePrompt.

For financial research, the analogous transformation is:

(9.1) RawMarketEvidence → FinanceTopologyIR → CandidateStructures → NumericValidationPlan + ResidualAudit.

9.3 Finance Topology Kernel

A reusable kernel for this paper can be written as:

```
Run as Equity Mediated-Excitation Topology Kernel.

Input:
  Stock universe, event window, price/volume data, factor residuals,
  attention traces, news/social text, ETF/fund overlap,
  options activity, ledger variables.

Protocol:
  Declare B = stock boundary.
  Declare Δ = observation rule.
  Declare h = emission, propagation, absorption, decay horizon.
  Declare u = admissible corrections.

Opcode sequence:
  Manifold → Coordinate → Boundary → Curvature → Flow → Gradient
  → Mediator → Coupling → Attractor → Bifurcation → Projection
  → Invariant → Holonomy → Residual → Compression → Phase-lock.

Output:
  1. Declared protocol.
  2. A-side excitation summary.
  3. Candidate mediator.
  4. Candidate absorber set.
  5. Coupling evidence.
  6. Rotation-force interpretation.
  7. Bifurcation gates.
```

```
8. Virtual / real trace test.
9. Alternative topology explanations.
10. Residual audit.
11. Required numerical validation.
```

This is not ornamental language. Each OpCode must correspond to a concrete operation.

The Runtime Kernel document states the discipline: no topology without operation, no compression without intent preservation, and no Kernel without residual audit.

9.4 OpCode interpretation for this paper

OpCode	Operation in mediated-transfer research
Manifold	define stock universe and state variables
Coordinate	choose return, volume, attention, liquidity, theme, ownership variables
Boundary	specify sector, theme, ETF, supply-chain, or event universe
Curvature	detect where simple factor model fails
Flow	identify capital, order, ETF, attention, or narrative flow
Gradient	estimate directional pressure from A to B
Mediator	identify narrative, liquidity, valuation, or memory carrier
Coupling	measure A-B relation under theme θ
Attractor	identify likely absorber or next carrier
Basin	define regime in which transfer logic holds
Bifurcation	identify gates that turn virtual heat into real trace
Projection	describe how observers reinterpret B through A's theme
Invariant	specify what must remain true across frames
Holonomy	test whether the loop A → theme → B → aftermath returns cleanly or leaves drift
Residual	report what remains unexplained
Compression	summarize the episode as a topology card
Phase-lock	detect whether observers synchronize around the same theme

This table is the bridge between theory and analysis.

The attached finance mapping already treats these OpCodes as a finance ontology: Manifold as market state-space, Boundary as mandate/law/liquidity/time horizon, Curvature as nonlinear tension, Flow as capital/cash/order/collateral flow, Gradient as directional pressure, Attractor as fair value/crowded trade/bubble/safe haven, and Residual as model error or hidden exposure.

9.5 Candidate topology structures

The LLM layer can generate topology cards such as:

Topology structure	Description
Leader-Emission Rotation	A becomes visible leader; theme transfers to B
Crowded Leader Exhaustion	A saturated; capital searches for unsaturated peer
Relative-Value Catch-Up	A-B valuation gap drives B response
Theme Basket Ignition	ETF or analyst basket propagates mediator
Squeeze Template Transfer	short-squeeze pattern moves across similar stocks
Trace-Gravity Revival	old leader reactivates because market remembers prior role
Liquidity Funnel	capital has few eligible routes and concentrates in B

Topology structure	Description
Gauge Reclassification	B is reclassified under a new theme
Failed Absorption	mediator reaches B, but damping blocks response
False Mediator	apparent transfer explained by common factor

Each structure should output:

```
Evidence:
Contrary evidence:
Required numerical test:
Ledger trace test:
Residual:
```

This reduces the risk of poetic overreach.

9.6 Multi-frame challenge

The same event should be analyzed under multiple observer frames.

Frame	Question
equity analyst	does B deserve rerating?
quant analyst	is response statistically significant?
liquidity trader	is flow route visible?
options trader	is gamma or call activity driving convexity?
credit analyst	does equity heat alter financing risk?
auditor / accounting frame	does the narrative match reported fundamentals?
regulator	is there manipulation or disorderly market behavior?
thematic fund manager	does B belong in the basket?
short seller	is the move unsupported by ledger trace?

A robust topology should survive frame challenge, or at least identify which frames disagree.

This corresponds to the finance OpCode document's point that finance is multi-frame: a position can appear safe under one frame but dangerous under another, and the purpose of OpCode analysis is to declare the financial world being analyzed rather than mixing layers unconsciously.

9.7 Residual audit

Every LLM-generated topology card must include residual audit.

Residual questions include:

- What data is missing?
- Could B have an independent catalyst?
- Could a common factor explain both A and B?
- Is the mediator identified before or after the response?
- Is the theme overfitted after the fact?
- Is the coupling measure stable out of sample?

- Is the ledger trace genuinely absent?
- Which observer frame would reject the interpretation?
- What would falsify this topology?

A usable residual audit may follow:

(9.2) ResidualAudit = MissingData + AlternativeExplanations + FrameDisagreement + TestFailureRisk + OverfitRisk.

This is essential because the framework introduces powerful vocabulary. Strong vocabulary without residual audit would create false sophistication.

9.8 Human-in-the-loop role

The LLM layer should not make final investment decisions.

Its role is:

(9.3) LLMRole = GenerateStructures + NameMediators + ChallengeFrames + AuditResiduals + ProposeTests.

The numerical layer's role is:

(9.4) NumericRole = MeasureEffects + EstimateParameters + TestHypotheses + ValidateRobustness.

The human analyst's role is:

(9.5) HumanRole = JudgeMateriality + ReviewEvidence + DecideAction + AcceptResponsibility.

This separation is important for governance.

The LLM is not the oracle. It is the topology compiler.

9.9 Why this may be a new research interface

The attached document explicitly notes that OpCode-style LLM use as a "Differential Topology Analysis Machine" is not established mainstream practice. It describes the synthesis as coming from the Kernel Skill architecture plus Gauge Grammar protocol-first finance logic, rather than from a known standard finance-AI product.

This matters because the proposal is not merely another sentiment model or factor screen.

It is a proposed research interface:

(9.6) Finance evidence → protocol declaration → topology compilation → candidate structure cards → numerical validation → residual audit.

This could be useful even when imperfect because it helps analysts see alternative structures before collapsing into a single explanation.

9.10 Summary of the LLM layer

The LLM OpCode Kernel is valuable if it satisfies:

(9.7) KernelValue = AlternativeCoverage + FrameDiversity + ResidualHonesty + Testability – DecorativeMetaphor – OverfitRisk.

It should be evaluated by whether it improves:

- structure discovery;
- scenario coverage;
- residual honesty;
- hypothesis generation;
- frame comparison;
- numerical test design.

Not by whether it predicts price directly.

The next sections should address limitations, anti-overreach rules, and the philosophical-financial interpretation of "force."

Transition to the Next Part

The next installment should cover:

Section 10 — Limitations and Anti-Overreach Rules
Section 11 — What Kind of "Force" Exists in Equity Markets?
Section 12 — Conclusion: From Market Narratives to Protocol-Compiled Regime Finance
Appendix A — OpCode-to-Finance Mapping Table

Draft Installment 4 — Sections 10–12 and Appendix A

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10. Limitations and Anti-Overreach Rules

10.1 Why this section is necessary

The framework proposed in this paper uses terms such as **Narrative Boson**, **Rotation Force**, **Virtual Excitation**, **Curvature**, **Attractor**, and **Ledger Trace**. These terms are useful only if they discipline financial analysis. They become dangerous if they merely decorate ordinary market commentary.

The framework must therefore obey a strict rule:

(10.1) Analytical value = diagnostic gain – metaphor risk – overfit risk – evidence gap.

The purpose is not to make markets sound like physics. The purpose is to make market-rotation phenomena more structured, testable, and auditable.

The attached finance OpCode document states this principle clearly: topology, gauge, and physics vocabulary is useful in finance only when it improves diagnosis, comparison, intervention, control,

or design; otherwise it should be removed.

This section therefore defines the anti-overreach rules required for the proposed theory.

10.2 Rule 1 — Do not claim literal quantum finance

The first rule is:

(10.2) Functional analogy ≠ physical identity.

A stock is not a particle. A theme is not a physical boson. A price movement is not a quantum event. A market is not literally a quantum vacuum.

The proper claim is:

(10.3) Some financial roles are structurally similar to field, mediator, gate, trace, residual, and force roles.

The term **Narrative Boson** means:

> a transferable market mediator that carries interpretive force from one stock to another.

It does not mean a real boson in the physical sense.

This distinction is necessary for credibility. The article's claim belongs to financial topology, behavioral finance, market microstructure, and semantic market dynamics—not to literal quantum physics.

10.3 Rule 2 — No protocol, no object

A claim about mediated transfer must begin with a declared protocol:

(10.4) $P = (B, \Delta, h, u)$.

Where B is the stock boundary, Δ is the observation rule, h is the time horizon, and u is the admissible correction set.

Without protocol, the object changes mid-analysis.

For example, suppose Stock A rises and Stock B rises later. The analyst may unconsciously switch between several frames:

Frame	Possible explanation
market frame	both moved because the market rose
sector frame	both moved because the sector rerated
factor frame	both are high-beta growth names
liquidity frame	both benefited from risk-on flow
theme frame	A emitted a theme that B absorbed
ETF frame	both are in the same basket
social frame	retail attention moved from A to B

All may be partially true. But the analyst must declare which financial world is being studied.

The finance OpCode document emphasizes this exact point: finance is not one object; it can be a price field, liquidity network, legal-contract system, accounting ledger system, narrative-expectation field, risk-transfer machine, or multi-observer collapse system.

Thus:

(10.5) No declared boundary → no stable transfer claim.

10.4 Rule 3 — No coupling measure, no force

A force claim requires a coupling measure.

It is not enough to say:

```
A rose; B is similar; therefore A caused B.
```

The framework requires:

(10.6) $g_{A,B,\theta}(t)$ = declared coupling under theme θ.

Coupling may be measured by:

- theme similarity;
- sector proximity;
- supply-chain linkage;
- ETF overlap;
- institutional co-ownership;
- analyst peer grouping;
- news co-mentions;
- social co-attention;
- options-flow similarity;
- historical activation.

If no coupling measure exists, the force language should not be used.

(10.7) $\text{ForceClaim}_{A\to B}$ is invalid if $g_{A,B,\theta}$ is undefined.

This prevents narrative overfitting.

10.5 Rule 4 — No mediator trace, no boson

A Narrative Boson requires evidence of mediation.

The analyst must identify what carried the relation from A to B.

Possible mediator traces include:

Mediator	Evidence

Mediator	Evidence
news mediator	theme appears in headlines linking A and B
analyst mediator	report identifies B as peer or beneficiary
social mediator	co-mentions and “next A” discourse
ETF mediator	shared basket flow
options mediator	related call activity
fund mediator	common holder rotation
screen mediator	similar high short-interest / low-float profile
supply-chain mediator	customer-supplier read-through

Without mediator evidence, the claim remains ordinary correlation.

(10.8) NarrativeBoson_A,θ is invalid if MediatorTrace_A,θ ≈ 0.

10.6 Rule 5 — No trace test, no virtuality

A no-trace price fluctuation must be tested against ledger variables.

The article’s virtuality claim is not:

```
The price rose and then fell.
```

It is:

```
The price rose, propagated, and decayed without durable ledger deformation.
```

Therefore:

(10.9) $\text{Virtual}_i \Leftrightarrow HL_i < K \wedge \|\Delta \text{Ledger}_i,t:t+H\| \leq \theta_L$.

If ledger variables are not tested, the event cannot be called virtual-like.

At most, one may say:

```
Candidate virtual excitation pending trace test.
```

This distinction is crucial. A price move may fade but still leave durable attention, ownership, or analyst trace.

10.7 Rule 6 — Do not erase real regime shifts through smoothing

A major risk of this framework is over-smoothing.

If the analyst aggressively removes market, sector, factor, and theme effects, the model may erase real structural change.

For example:

- a technology platform genuinely enters a new demand regime;
- a bank genuinely loses funding access;
- a defence company genuinely receives a durable order cycle;

- an energy firm genuinely benefits from supply scarcity;
- a biotech firm genuinely changes its probability distribution after trial results.

These are not virtual heat. They are real ledger-changing events.

Therefore:

(10.10) Smoothing must not remove gate events.

A gate event is:

(10.11) $Gate_{i},t = 1$ if event changes future admissible path.

The correct question is not “can we smooth the chart?” but:

(10.12) Which part of the movement is common-field adjustment, which part is virtual excitation, and which part is ledger-changing transition?

10.8 Rule 7 — Avoid look-ahead bias

All factor adjustments must use information available before the event.

Correct:

(10.13) $\beta_{i},t-1 = \text{Estimate}(\beta_{i} \mid \text{data} \le t-1)$.

Incorrect:

(10.14) $\beta_{i} = \text{Estimate}(\beta_{i} \mid \text{full sample})$.

Using full-sample beta to interpret past events creates causal contamination. It allows future structure to explain past behavior.

This is especially dangerous in a theory of self-reference, because the market’s future interpretation may have changed precisely because of the event being studied.

10.9 Rule 8 — Beware endogenous market adjustment

The market index is not a neutral external field.

In equity markets, the broad index is itself a recursive expectation aggregate. If a sector is large enough, its own excitation may influence the market factor used to adjust it.

Thus:

(10.15) r_market may contain part of r_sector.

To reduce self-cancellation:

(10.16) r_m,ex-s,t = market return excluding sector s.

For very large stocks, even ex-sector adjustment may not fully remove endogeneity because index movement, ETF flow, and macro sentiment may be co-created by the leader stock itself.

Thus:

(10.17) Market correction is a gauge choice, not an absolute truth.

10.10 Rule 9 — Separate hypothesis generation from validation

LLM topology analysis should not be treated as proof.

The LLM can help generate candidate structures:

- “Liquidity Funnel”
- “Narrative Phase-Lock”
- “Crowded Leader Exhaustion”
- “Relative-Value Catch-Up”
- “Trace-Gravity Revival”
- “Failed Absorption”

But each named topology must output measurable validation hooks.

The attached document makes the same methodological point: numeric methods discover measured patterns, while LLM OpCodes compile semantic-financial structures; together they produce topology-aware finance diagnosis.

Therefore:

(10.18) LLMTopologyClaim is admissible only if it produces evidence tags and numeric tests.

Or more compactly:

(10.19) No test → weak topology.

10.11 Rule 10 — Evidence tags are mandatory

Every topology claim should be tagged.

Suggested evidence tags:

Tag	Meaning
data-supported	supported by price, volume, flow, or factor data
text-supported	supported by news, reports, filings, transcripts
institution-supported	supported by ETF, ownership, index, or legal structure
inferred	plausible but not directly observed
speculative	weakly supported hypothesis
residual	unresolved but material
contradicted	evidence against the interpretation

This mirrors the attached document’s recommendation that every topology claim be tagged as supported by data, supported by text, inferred, speculative, residual, or unknown.

The purpose is to prevent the vocabulary from becoming overconfident.

10.12 Rule 11 — Multi-frame disagreement must be preserved

A mediated-transfer event may look different under different observer frames.

Observer frame	Likely interpretation
momentum trader	continuation or rotation signal
value investor	overreaction or relative mispricing
short seller	unsupported hype
credit analyst	irrelevant unless financing changes
options trader	volatility and convexity event
ETF manager	basket flow effect
regulator	possible disorderly trading or manipulation
company management	opportunity for communication or financing
auditor	no effect unless accounting or disclosure changes

A strong analysis should not collapse these frames prematurely.

(10.20) FrameDisagreement_P = {Interpretation_j – Interpretation_k | observers j,k under protocol P}.

If frame disagreement is high, the result should be marked as unstable.

This follows the OpCode framework's emphasis on multi-frame interpretation: finance changes radically by frame, and the same entity can look safe under accounting capital but fragile under liquidity or depositor psychology.

---

11. What Kind of "Force" Exists in Equity Markets?

11.1 The central question

If Stock A's excitation can increase Stock B's abnormal response probability through a mediator, what kind of "force" is this?

It is not a physical force.

It is not a legal force.

It is not merely money flow.

It is not merely correlation.

It is a directional pressure in the attention-capital manifold.

The proposed definition is:

(11.1) MarketForce_A→B,θ = directional pressure created when A's visible trace is compressed into mediator θ and projected onto B through coupling g_A,B,θ.

In plain language:

> A force exists when the market's interpretation of A changes the probability that capital

and attention move into B.

This is why the force is observer-mediated.

Without observers, price traces do not become themes. Without themes, A does not become a search template. Without search, B is not reclassified. Without reclassification, capital does not rotate.

Thus:

(11.2) Force = ProjectionGradient in the attention-capital manifold.

11.2 The force is not money itself

Money is the medium of transaction, but not necessarily the cause of direction.

Capital does not flow randomly. It flows through interpretable routes.

A stock becomes a destination when it is made legible as:

- the next beneficiary;
- the cheaper peer;
- the forgotten leader;
- the short-squeeze candidate;
- the ETF constituent;
- the supply-chain read-through;
- the policy winner;
- the risk proxy;
- the recovery candidate.

The force is the directional structure that makes one destination more likely than another.

Thus:

(11.3) Flow = movement of capital.

(11.4) Force = reasoned pressure that gives flow a direction.

The reason may be rational, semi-rational, reflexive, or purely mimetic. The framework does not assume that the force is fundamentally correct. It only asks whether it is measurable.

11.3 Force family 1 — Attention Gradient Force

The first force family is attention gradient.

When A becomes visible, observers search for related assets. B rises if it is both related and previously under-observed.

(11.5) $F_{attention,A\to B} = AttentionShock_A \cdot Similarity_{A,B} \cdot VisibilityGap_B$.

Where:

(11.6) VisibilityGap_B = PotentialVisibility_B – CurrentVisibility_B.

A high attention force means:

- A has become highly visible;
- B is similar enough to be found;
- B has not yet received equivalent attention.

This force often appears in thematic rallies, meme rotations, and retail-driven markets.

Its virtual form is common: B receives attention, rises briefly, and fades.

Its real form occurs when new attention causes durable analyst coverage, institutional ownership, or financing access.

11.4 Force family 2 — Narrative Force

Narrative force arises when a story becomes transferable.

(11.7) F_narrative,A→B,θ = ThemeFit_B,θ · Emit_A,θ · Receptivity_θ – NarrativeDamping_B.

A story is not equally transferable to every stock.

It requires:

- theme fit;
- investor receptivity;
- credible mapping;
- low contradiction;
- enough liquidity;
- enough imagination.

Narrative force is strongest when the market can say:

```
If A benefits from θ, B may also benefit from θ.
```

It is weaker when the mapping requires too many assumptions.

Narrative force is the most direct expression of the Narrative Boson model.

11.5 Force family 3 — Relative-Value Force

Relative-value force arises when A's move creates valuation tension with B.

(11.8) F_RV,A→B = ValuationGap_A,B · FundamentalSimilarity_A,B · InvestorCapacity_B.

A rises. B does not. If A and B are perceived as similar, B becomes cheap relative to A.

This creates catch-up pressure.

The key distinction is:

(11.9) Relative-value force may be rational even when narrative force is mimetic.

However, relative-value force can also become reflexive. If investors buy B because it has not yet moved, the act of buying validates the catch-up thesis.

11.6 Force family 4 — Liquidity Routing Force

Liquidity routing force arises from the institutional plumbing of markets.

(11.10) $F_{liquidity,A\to B} = BasketRoute_{A,B} \cdot FlowIntensity_{\theta} \cdot Tradability_B - LiquidityFriction_B$.

This force travels through:

- ETFs;
- mutual funds;
- index products;
- thematic baskets;
- broker baskets;
- quantitative screens;
- derivatives hedging;
- risk-parity or volatility-targeting systems.

A theme may become investable only through a basket. If B is inside that basket, it may receive flow even before discretionary investors form a strong opinion.

This is closer to an institutional exchange channel than a pure narrative channel.

11.7 Force family 5 — Trace-Gravity Force

Trace-gravity force arises from market memory.

Some stocks have previously served as leaders, meme objects, crisis proxies, or thematic carriers. Their past excitations leave memory. When similar conditions return, the market bends back toward them.

(11.11) $F_{trace,B} = PastExcitationMemory_B \cdot CurrentThemeResonance_B \cdot ObserverRecall_B$.

This explains why old leaders often revive quickly.

The stock's past role becomes a gravitational well in the market's memory.

Examples:

- former AI leaders during renewed AI heat;

- old meme stocks during retail-squeeze episodes;
- previous crisis banks during renewed banking stress;
- old commodity winners during commodity spikes.

This force may have no direct new fundamental cause. Its mediator is memory.

11.8 Force family 6 — Gauge-Reclassification Force

Gauge-reclassification force occurs when the market changes the frame under which B is interpreted.

A stock previously viewed as an industrial company may be reclassified as an AI infrastructure company. A car company may be reclassified as a robotics or energy company. A software company may be reclassified as a cybersecurity company.

(11.12) $F_{gauge,B} = \Delta Frame_B \cdot CapitalAvailable_{under_new_frame} - OldFrameDamping_B$.

The force arises from a change in category.

Under the old frame, B's valuation multiple may be limited. Under the new frame, B may enter a different peer group, different investor base, and different narrative basin.

Gauge reclassification is powerful because it changes not merely the price but the coordinate system.

In OpCode language:

(11.13) GaugeChange = change of local representation that preserves some identity while altering admissible interpretation.

This is why a gauge-reclassification event can become real if it enters the analyst, ownership, and valuation ledger.

11.9 Force family 7 — Phase-Lock Force

Phase-lock force appears when many observers synchronize around the same theme.

(11.14) $F_{phase,\theta} = ObserverAgreement_\theta \cdot FlowSynchronization_\theta \cdot Repetition_\theta$.

When phase-lock is high:

- headlines repeat the same theme;
- analysts publish similar beneficiary lists;
- funds buy the same baskets;
- algorithms detect the same momentum;
- social platforms repeat the same tickers;
- options activity clusters around the same names.

The attached finance OpCode mapping treats phase-lock as herding, index rebalancing, central-

bank narrative alignment, or synchronized selling.

Phase-lock can create rapid transfer, but also fragility. If the shared frame breaks, reversal may also become synchronized.

11.10 A unified force expression

The combined rotation force may be written as:

(11.15) $F_{A\to B,\theta} = F_{attention} + F_{narrative} + F_{RV} + F_{liquidity} + F_{trace} + F_{gauge} + F_{phase} - D_B$.

Or:

(11.16) $F_{A\to B,\theta} = \Sigma_m F_{m,A\to B,\theta} - D_B$.

Where m indexes force families.

The force becomes tradable when:

(11.17) $F_{A\to B,\theta} > \Theta_B$.

Where Θ_B is B's activation threshold.

A stock with high liquidity, high theme fit, strong observer overlap, and low saturation has a lower activation threshold. A stock with high damping, low credibility, high valuation, or weak liquidity has a higher threshold.

11.11 Force, Peak, and Trap

Force can produce different outcomes.

If force is temporary and no ledger gate is crossed:

(11.18) $F_{A\to B} > \Theta_B \to VirtualHeat_B \to Decay$.

If force crosses a ledger gate:

(11.19) $F_{A\to B} > \Theta_B \wedge Gate_B = 1 \to RealRerating_B$.

If force becomes self-validating:

(11.20) $\partial Demand_B / \partial p_B > 0 \to Peak_B$.

If negative force closes future access:

(11.21) $NegativeShock_B \to Access_B\downarrow \to Expectation_B\downarrow \to NegativeShock_B$.

This is Trap formation.

The Peaks and Traps framework supports this logic by arguing that self-reference makes price both a parameter and an expectation-fixed point, producing positive-slope demand or closed-access basins when recursive expectations dominate.

Thus, the proposed force model can be integrated into a broader theory of self-referential finance.

11.12 The deepest interpretation

The deepest interpretation is:

> Equity-market force is the directional curvature created when observer attention, tradable narrative, institutional routing, and capital capacity align around a possible future.

In compact form:

(11.22) Force = ∇_B ExpectedProjectionValue(B | A excitation, θ, P).

The force pulls not toward intrinsic truth, but toward projected tradability.

This is why the same stock can remain dormant under one frame and suddenly move under another. The stock did not change first. The projection field changed first.

That is the central insight of mediated excitation transfer.

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12. Conclusion — From Market Narratives to Protocol-Compiled Regime Finance

12.1 Summary of the argument

This paper proposed a protocol-topological theory of mediated excitation transfer in equity markets.

The core claim is that some equity-market movements are not well described as isolated stock-specific shocks or random residuals. Instead, they are mediated transfers. Stock A becomes excited. Its visible trace is compressed into a tradable mediator. That mediator propagates through a similarity, ownership, liquidity, or attention graph. Stock B absorbs it. B's abnormal response may then decay without trace or pass through a ledger gate and become real.

The main sequence is:

(12.1) A excitation → mediator formation → B absorption → price response → trace test.

The central mediator is the **Narrative Boson**:

(12.2) $NB_{A,\theta,t}$ = Compress($Trace_{A,t}$ | θ, P).

The central force is the **Observer-Mediated Rotation Force**:

(12.3) $F_{A\to B,\theta}(t,k) = g_{A,B,\theta}(t) \cdot E_A(t) \cdot U_B(t) \cdot L_B(t) \cdot M_{A,B,\theta}(t) - D_B(t)$.

The central classification is:

(12.4) VirtualTransfer$_{A\to B} \Leftrightarrow \varepsilon_{B,t+k} \neq 0 \wedge \Delta Ledger_B \approx 0$.

(12.5) RealTransfer$_{A\to B} \Leftrightarrow \varepsilon_{B,t+k} \neq 0 \wedge \Delta Ledger_B \neq 0$.

This gives a way to distinguish tradable but transient heat from durable regime change.

12.2 The methodological contribution

The paper's methodological contribution is not merely the vocabulary of Narrative Bosons or Rotation Forces. The deeper contribution is the protocol structure.

Every transfer claim must declare:

(12.6) $P = (B, \Delta, h, u)$.

Where B is boundary, Δ is observation rule, h is horizon, and u is admissible correction.

This prevents uncontrolled frame switching. It also makes the theory empirically testable.

A valid claim requires:

(12.7) $\text{ValidTransferClaim}_P \Leftrightarrow \text{ProtocolDeclared} \wedge \text{ExcitationDetected} \wedge \text{MediatorIdentified} \wedge \text{CouplingMeasured} \wedge \text{ResponseTested} \wedge \text{ResidualAudited}$.

This is the difference between a market story and a research framework.

12.3 The empirical contribution

The empirical design proposed here includes:

1. Define a stock universe.
2. Build a similarity graph.
3. Remove market, sector, factor, liquidity, and event effects.
4. Detect A-side excitation.
5. Identify mediator trace.
6. Measure B response.
7. Compare high-coupling pairs against placebo pairs.
8. Test ledger persistence.
9. Classify the topology card.
10. Audit residual.

This allows the analyst to test whether a rotation episode is:

- virtual heat;
- technical residue;
- hidden trace;
- real rerating;
- Peak;
- Trap;
- failed absorption;
- false mediator.

This structure turns vague market language into analyzable market topology.

12.4 The AI contribution

The paper also proposed an LLM OpCode Kernel as a qualitative topology compiler.

The role of the LLM is not to forecast price. It is to compile evidence into candidate structures, name possible mediators, compare observer frames, and audit residual uncertainty.

The pipeline is:

(12.8) RawMarketEvidence → ProtocolTrace → RoleMap → TopologyCards → NumericValidation → ResidualAudit.

This matches the attached document's view that a useful finance topology workbench should include protocol declaration, evidence map, numeric indicators, candidate topology cards, cross-frame disagreement, bifurcation watchlist, invariant checks, residual audit, required numeric validation, and human decision notes.

The LLM layer is therefore:

(12.9) LLMRole = GenerateStructures + NameMediators + ChallengeFrames + AuditResiduals + ProposeTests.

The numerical layer is:

(12.10) NumericRole = MeasureEffects + EstimateParameters + TestHypotheses + ValidateRobustness.

The human analyst remains responsible for:

(12.11) HumanRole = JudgeMateriality + ReviewEvidence + DecideAction + AcceptResponsibility.

12.5 Theoretical contribution

The theoretical contribution is a new interpretation of equity-market "force."

In this framework:

(12.12) Force = ProjectionGradient in the attention-capital manifold.

More fully:

(12.13) Market force is the directional pressure created when observer attention, tradable narrative, institutional routing, and capital capacity align around a possible future.

This force is not physical. It is financial, semantic, institutional, and behavioral.

It may appear as:

- attention force;
- narrative force;
- relative-value force;

- liquidity-routing force;
- trace-gravity force;
- gauge-reclassification force;
- phase-lock force.

These forces are not mutually exclusive. A strong market episode may contain several at once.

For example, an AI leader rally may involve:

```
attention force + narrative force + ETF liquidity force + relative-value force + phase-lock force.
```

A former leader revival may involve:

```
trace-gravity force + narrative force + attention force.
```

A thematic bubble may involve:

```
narrative force + phase-lock force + positive-slope demand.
```

A funding crisis may involve:

```
negative attention force + liquidity-routing force + trap dynamics.
```

This vocabulary gives analysts a way to classify market regimes more precisely.

12.6 Relation to random walk thinking

This paper does not deny random-walk behavior.

Instead, it refines the object.

Raw stock prices may look random-walk-like because they contain:

- market drift;
- sector drift;
- factor exposure;
- liquidity shocks;
- firm-specific events;
- narrative heat;
- institutional flow;
- observer self-reference;
- residual noise.

After protocol-declared adjustment, some residuals may still behave like random walks. Others may display short-lived no-trace excitations. Others may reveal real regime transitions.

Thus:

(12.14) RandomWalkAppearance = ExtractableStructure_T^c + Residual_T under observer

bound T.

A bounded observer may see random walk because the structure is not extractable under its current protocol.

The question is not whether the market is random or deterministic. The question is:

(12.15) Under which protocol does mediated structure become visible?

12.7 Practical implications

The proposed framework can support:

Use case	Contribution
equity research	classify rotation episodes and peer read-throughs
quant strategy	test A-to-B residual propagation
thematic investing	identify real vs virtual theme transfer
risk management	detect crowded phase-lock and false mediators
short selling	find no-trace heat vulnerable to decay
portfolio construction	avoid overexposure to one narrative boson
options strategy	identify convex absorption candidates
market surveillance	distinguish organic rotation from manipulation-like propagation
AI finance tools	generate topology cards and residual audits

The strongest immediate product is not an autonomous trading system. It is a **Finance Topology Workbench**.

Its output should be:

1. protocol declaration;
2. evidence map;
3. excitation summary;
4. mediator candidate;
5. coupling graph;
6. absorber set;
7. rotation-force estimate;
8. virtual / real trace test;
9. cross-frame disagreement;
10. residual audit;
11. numerical validation plan.

12.8 Final thesis

The final thesis of the paper is:

(12.16) Equity markets contain mediated self-referential excitations that can be studied as protocol-bound transfers across a stock manifold.

And:

(12.17) A Narrative Boson is a transferable market mediator formed when observers compress one stock's visible excitation into a tradable theme.

And:

(12.18) A Rotation Force is the directional pressure by which that mediator raises another stock's abnormal response probability.

And:

(12.19) A No-Trace Price Fluctuation is a mediated response that decays without durable ledger deformation.

The contribution is not to replace existing finance. It is to add a structured layer above raw price action and below broad market narrative.

In one sentence:

> This paper turns the phrase "the heat rotated to another stock" into a protocol-declared, factor-adjusted, mediator-tested, ledger-audited research object.

Appendix A — OpCode-to-Finance Mapping for Mediated Excitation Transfer

OpCode	General operation	Finance mapping	Use in this paper
Kernel	establish runtime analysis identity	investment thesis / risk engine / topology model	defines the mediated-transfer analysis mode
Manifold	define multidimensional state space	price, volume, liquidity, attention, theme, ownership, factors	constructs the stock manifold
Coordinate	identify variables	residual return, volume surprise, attention, ETF overlap, theme vector	selects measurable axes
Chart	create local representation	equity view, sector view, theme view, liquidity view	prevents frame confusion
Boundary	define scope and constraints	stock universe, sector, ETF basket, theme cluster	declares eligible emitters and absorbers
Curvature	detect nonlinear tension	residual after factor correction; valuation-liquidity contradiction	identifies where simple models fail
Flow	describe movement	capital flow, order flow, ETF flow, attention flow, narrative flow	tracks propagation channels
Gradient	identify direction of change	rotation pressure toward B	defines force direction
Mediator	carrier of interaction	narrative boson, liquidity route, valuation signal	explains A-to-B transfer

OpCode	General operation	Finance mapping	Use in this paper
Coupling	measure connection	theme similarity, ETF overlap, co-ownership, co-mentions	estimates g_A,B,θ
Attractor	stable convergence zone	next carrier, crowded trade, old leader, safe haven	identifies likely absorber
Basin	scope of attractor validity	regime in which theme transfer works	prevents overgeneralization
Bifurcation	future-branching event	earnings, guidance, ETF inclusion, policy, financing	tests virtual-to-real transition
Singularity	model breakdown	trading halt, default, squeeze explosion, liquidity freeze	marks failure of ordinary model
Projection	convert high-dimensional state into visible output	price chart, analyst model, headline, ranking screen	explains observer collapse into trade idea
Invariant	preserve non-negotiable relation	accounting identity, no-arbitrage relation, solvency condition	checks that interpretation does not violate basics
Holonomy	loop consistency after round trip	A → theme → B → aftermath; forecast → actual → revised forecast	detects hidden drift
Residual	unresolved remainder	unexplained return, hidden leverage, untested mediator, missing data	audit layer
Compression	reduce complexity while preserving structure	topology card, factor model, research note	produces usable summary
Phase-lock	alignment among agents	herding, synchronized buying, repeated narrative, basket flow	detects self-reinforcing market rhythm

This mapping is aligned with the attached finance OpCode document, which treats the finance mapping as a multi-perspective financial analysis compiler and lists finance equivalents for Manifold, Coordinate, Boundary, Curvature, Flow, Gradient, Attractor, Bifurcation, Projection, Invariant, Residual, Compression, and Phase-lock.

Appendix B — Episode Topology Card Template

```
Episode Name:

Protocol:
B = boundary / stock universe
Δ = observation rule
h = emission, propagation, absorption, decay horizon
u = admissible corrections

Emitter:
Stock A:
Excitation date:
A-side excitation score:
A-side trace evidence:

Candidate Mediator:
Theme θ:
Narrative boson evidence:
Mediator channels:
Emission strength:
```

```
Absorber:
Stock B:
Coupling g_A,B,θ:
Unsaturation U_B:
Liquidity L_B:
Damping D_B:

Rotation Force:
Estimated F_A→B,θ:
Expected lag k:
Observed B response:
Placebo comparison:

Trace Test:
Price half-life:
Ledger variables checked:
Ledger persistence result:
Virtual / real classification:

Alternative Explanations:
Common factor:
Direct B news:
Sector move:
ETF flow:
Options channel:
Other:

Residual Audit:
Missing data:
Weak assumptions:
Frame disagreements:
Required validation:
```

Appendix C — Minimal LLM Finance Topology Kernel

```
Run as Equity Mediated-Excitation Topology Kernel.

Purpose:
Analyze whether Stock A's abnormal excitation transferred to Stock B through a mediator.

Protocol:
Declare B = stock universe.
Declare Δ = observation rule.
Declare h = emission / propagation / absorption / decay horizon.
Declare u = admissible corrections.

Opcode Stack:
Manifold → Coordinate → Boundary → Curvature → Flow → Gradient
→ Mediator → Coupling → Attractor → Bifurcation → Projection
→ Invariant → Residual → Compression → Phase-lock.

Tasks:
1. Identify A-side excitation.
2. Identify candidate mediator.
3. Define A-B coupling.
4. Estimate B absorption conditions.
5. Describe rotation force.
6. Test virtual vs real trace.
7. List alternative explanations.
8. Audit residual.
9. Propose numeric validation.

Rules:
Do not predict price directly.
```

```
Do not invent facts.
Separate data-supported, text-supported, inferred, speculative, and residual claims.
No coupling measure, no force.
No mediator trace, no boson.
No ledger test, no virtuality.
Always report residual.
```

Appendix D — Formula Summary

(0.1) Equity-market heat does not merely diffuse; it can be mediated, compressed, transferred, absorbed, damped, and ledger-tested.

(1.1) $r_{i,t} = \alpha_{i,t-1} + \beta_{i,t-1} r_{m,t} + \beta^{s}_{i,t-1} r^{s}_{,t} + \beta^{f}_{i,t-1} F_t + \varepsilon_{i,t}$.

(2.1) $\mathcal{M}_P$ = {Price, Volume, Volatility, Attention, Liquidity, Theme, Ownership, Options, Fundamentals, Risk | P}.

(2.4) $E_i(t) = w_1 z(\varepsilon_{i,t}) + w_2 z(\text{VolumeSurprise}_{i,t}) + w_3 z(\text{Attention}_{i,t}) + w_4 z(\text{OptionActivity}_{i,t})$.

(2.6) $NB_A,\theta,t = \text{Compress}(\text{Trace}_A,t \mid \theta, P)$.

(2.8) $\text{Absorb}_B(NB_A,\theta,t) \propto g_A,B,\theta(t) \cdot U_B(t) \cdot L_B(t) \cdot V_B(t)$.

(2.9) $F_A \to B,\theta(t,k) = g_A,B,\theta(t) \cdot E_A(t) \cdot U_B(t) \cdot L_B(t) \cdot M_A,B,\theta(t) - D_B(t)$.

(2.11) $\text{Virtual}_B \Leftrightarrow \Delta p_B \neq 0 \wedge \Delta\text{Ledger}_B \approx 0$.

(2.12) $\text{Real}_B \Leftrightarrow \Delta p_B \neq 0 \wedge \Delta\text{Ledger}_B \neq 0$.

(3.1) $P = (B, \Delta, h, u)$.

(3.14) $\text{ValidTransferClaim}_P \Leftrightarrow \text{ProtocolDeclared} \wedge \text{ExcitationDetected} \wedge \text{MediatorIdentified} \wedge \text{CouplingMeasured} \wedge \text{ResponseTested} \wedge \text{ResidualAudited}$.

(7.8) $\text{Virtual}_i \Leftrightarrow HL_i < K \wedge \|\Delta\text{Ledger}_{i,t:t+H}\| \leq \theta_L$.

(7.9) $\text{Real}_i \Leftrightarrow HL_i \geq K \vee \|\Delta\text{Ledger}_{i,t:t+H}\| > \theta_L$.

(7.11) $VR_P = N_\text{virtual} / N_\text{total}$.

(8.16) $\varepsilon_B,t+k = c + \beta_F F_A \to B,\theta(t,k) + \text{Controls} + \eta_B,t+k$.

(10.1) Analytical value = diagnostic gain − metaphor risk − overfit risk − evidence gap.

(11.2) Force = ProjectionGradient in the attention-capital manifold.

(11.16) $F_A \to B,\theta = \Sigma_m F_m,A \to B,\theta - D_B$.

(12.7) $\text{ValidTransferClaim}_P \Leftrightarrow \text{ProtocolDeclared} \wedge \text{ExcitationDetected} \wedge \text{MediatorIdentified} \wedge \text{CouplingMeasured} \wedge \text{ResponseTested} \wedge \text{ResidualAudited}$.

End of Main Draft

This completes the full first-draft article.

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This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

I am merely a midwife of knowledge.